Technical Environmental Diary

A technical environmental diary compiled across the Climate, Energy and Architecture module, recording lectures, fieldwork, building visits and design research on daylighting, thermal comfort, natural ventilation, passive heating and energy generation.



The diary tracks a year of environmental design learning, from climate and site analysis through human comfort, passive design strategies, energy systems and industry building visits, applying each topic back into live design work.

01/09/2018 - date started

01/06/2019 - date ended



Documents by Sebastian Dawber
https://www.sebastianvictordawber.co

Created for ...
eg BSc Architecuture and enviromental design
Undergraduate RIBA ARB part I

All work produced by Sebastian Victor Dawber

---

https://www.sebastianvictordawber.co/

---

This is work produced by the parties mentioned anyone using the work without agreement signiture and physical represented has done so in their own doing without the parties mentioned

---

Https://www.instagram.com/sarc_studio/

---



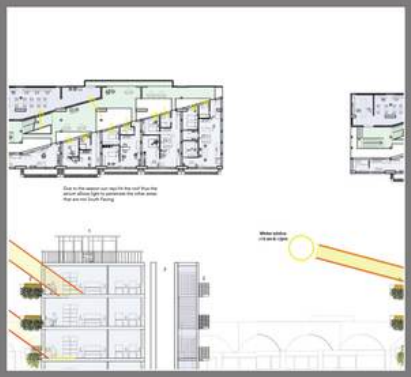
SEBASTIAN VICTOR DAWBER
Year 2
BSc Part 1, ARB RIBA , Architecture and environmental design

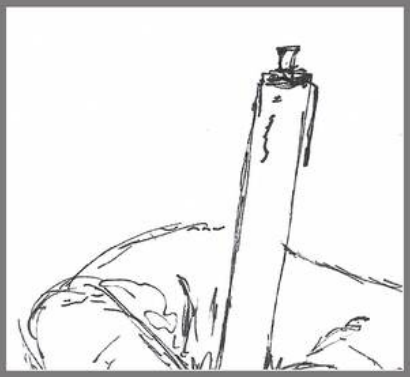
# Technical Environmental Diary

## Climate, Energy & Architecture Logbook

This document is a technical environmental diary compiled over the Climate, Energy and Architecture module at the University of Westminster. It records two semesters of lectures and fieldwork, covering climate and site analysis, human comfort, passive heating and natural cooling, natural ventilation, daylighting and thermal comfort, energy generation and water systems. The diary also documents building visits, including Saint-Gobain, Future Build, the Canary Wharf Crossrail roof garden and the Crystal, translating lessons from industry into design research such as atrium ventilation, daylight modelling and urban hydroponics. Together these entries form a running record of how environmental principles were tested and applied through the year.

01/06/2019 - date ended







# Technical enviromental diary
# By Sebastian Dawber
-TES2
- W16104965

- University of Westminster school of Architecture and cities
- BSC Architecture and enviromental design
- 2018/2019: Climate, energy and Architecture module

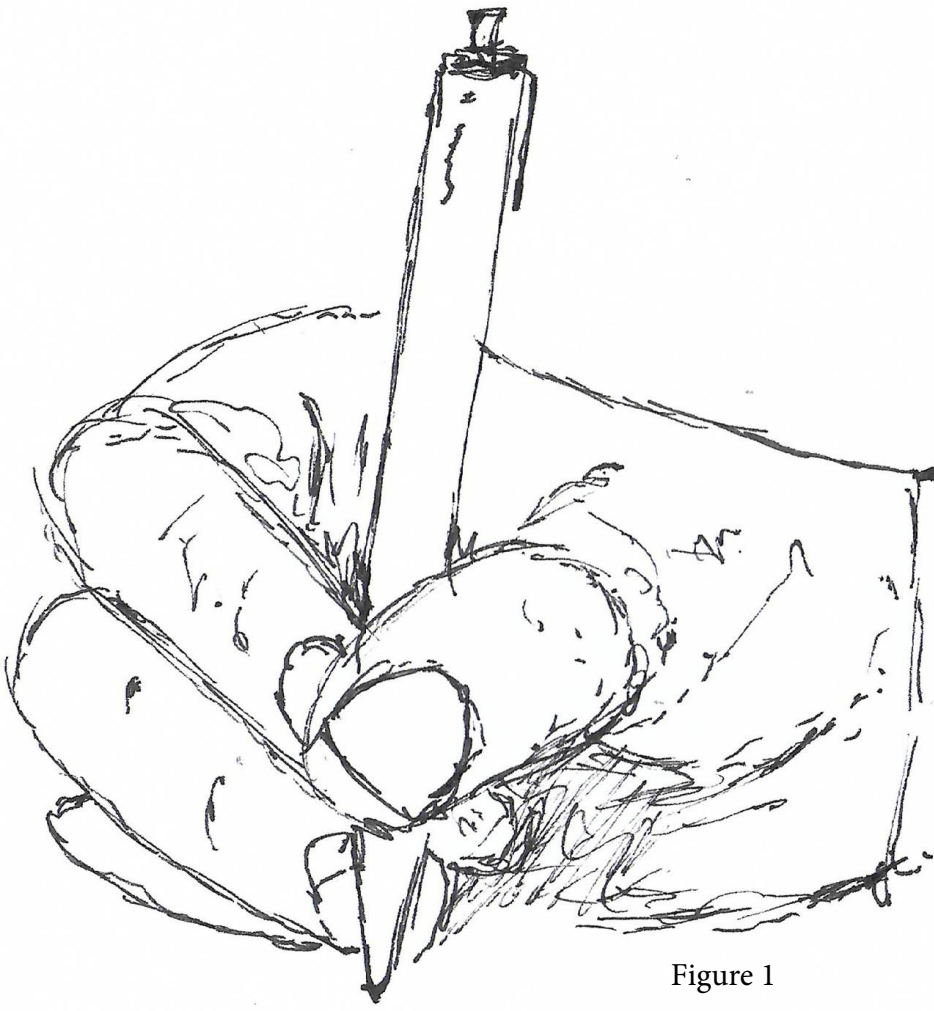


Figure 1

# Contents page

Semester 1


- Lecture 1- Introduction
- Lecture 2 – Climate Analysis
- Lecture 3 – Site Analysis + Fieldwork
- Lecture 4 – Human Comfort
- Lecture 5– Building Analysis + Fieldwork
- Lecture 6 – Passive Heating
- Lecture 7 – Indoor Temperature Calculator
- Lecture 8 – Natural Ventilation
- Lecture 9 – Optivent Tool
- Lecture 10 – Natural Cooling
- Lecture 11 – Building Envelope
- End of semster 1


Semester 2


- Lecture 12- Daylighting: designing for solar control
- Lecture 13– Thermal comfort
- Lecture 14– Thermal comfort workshop
- Lecture 15 – Energy generation
- Lecture 16– Building visit: Saint Gobain
- Lecture 17 – Building visit: Future build
- Lecture 18 – Building visit: Canary Wharf Crossrail roof garden
- Lecture 19 – Codes & regulations
- Lecture 20– Water systems
- Lecture 21 – Building visit: the Crystal




# Introduction

The aim of this module is to describe the dynamic relationship between external and internal environments and explore how the design, materials construction and specification can successfully mediate our built environment.

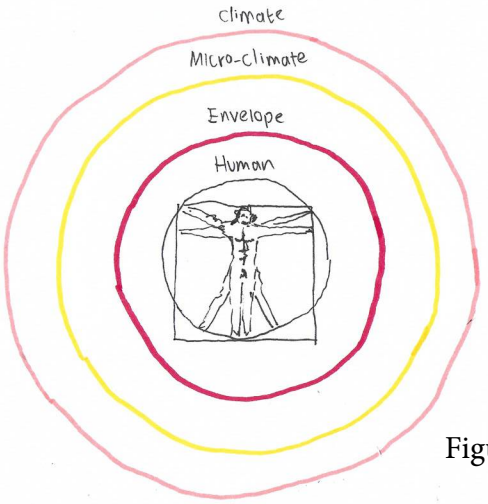



Figure 2

## Aims of the TES2 module

o To explore analytical tools with which to study, record and annotate local environmental factors such as sunlight, daylight, temperature, humidity, precipitation and sound.

o To collaborate in a group to conduct fieldwork and develop design strategies for the passive heating and cooling of buildings.

o To present the outcome of analytical design studies in an illustrated Technical Design Diary and Site Analysis Report.

o To understand the contemporary context of passive technologies and design strategies that we can be employed to heat, cool



Figure 3

# Climate analysis

Looking at the Golden lane site we started to learn how to do an enviromental analysis

- Effects of microclimate: surfaces, materials and geometry shapes
- Wind- unstable and has many variables effecting it e.g. air vortexes and movements
- Cities are warmer at night as they release heat collected during the day
- Calibrating data taken using the various instruments, use a benchmark to test how reliable the data is
- Pollution- types vary  trees can be used to combat pollution

Climate: The weather conditions prevailing in an area in general or over a long period. The wolrds has various climatic zones that effect each reigon in a different way.

Micro-climate: the climate of a very small or restricted area, especially when this differs from the climate of the surrounding area

A - Tropical Moist Climates: all months have average temperatures above 18° Celsius.

B - Dry Climates: with deficient precipitation during most of the year.

C - Moist Mid-latitude Climates with Mild Winters.

D - Moist Mid-Latitude Climates with Cold Winters.

E - Polar Climates: with extremely cold winters and summers.

## The Köppen climate classification :by the Russian climatologist Wladimir Köppen

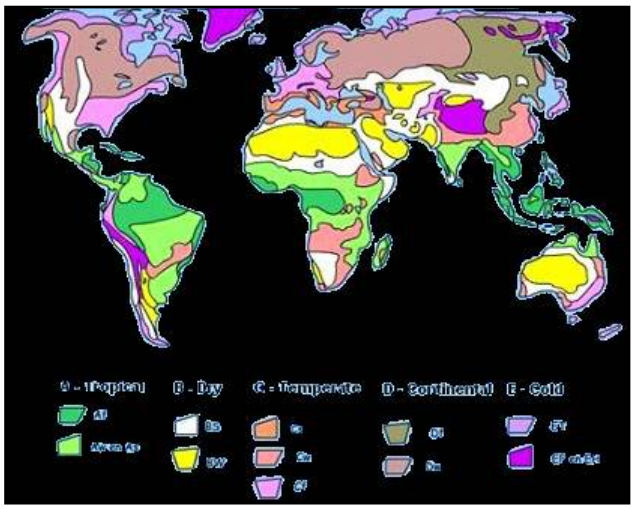



Figure 4

- Group A: Tropical (megathermal) climates

-This type of climate has every month of the year with an average temperature of 18 °C or higher, with significant precipitation.

- Group B: Dry (arid and semiarid) climate

-This type of climate is defined by little precipitation.

- Group C: Temperate (mesothermal) climates

This type of climate has the coldest month averaging between 0 °C or  −3 °C and 18 °C and at least one month averaging above 10 °C

- Group D: Continental (microthermal) climates

Rhis type of climate has at least one month averaging below 0 °C or −3 °C and at least one month averaging above 10 °C .

- Group E: Polar and alpine (montane) climates

This type of climate has every month of the year with an average temperature below 10 °C .

# Design relation

This analysis allowed me to undersatnd how to conduct a microclmiate analysis with the various instruments. Allowing me to test various varibales that effect desing the core elements to enviromental design. Using the understanding of the climate, in this case the United kingdom; to analyse the Golden Lane estate to help us improve any design that may be implamented in that area.

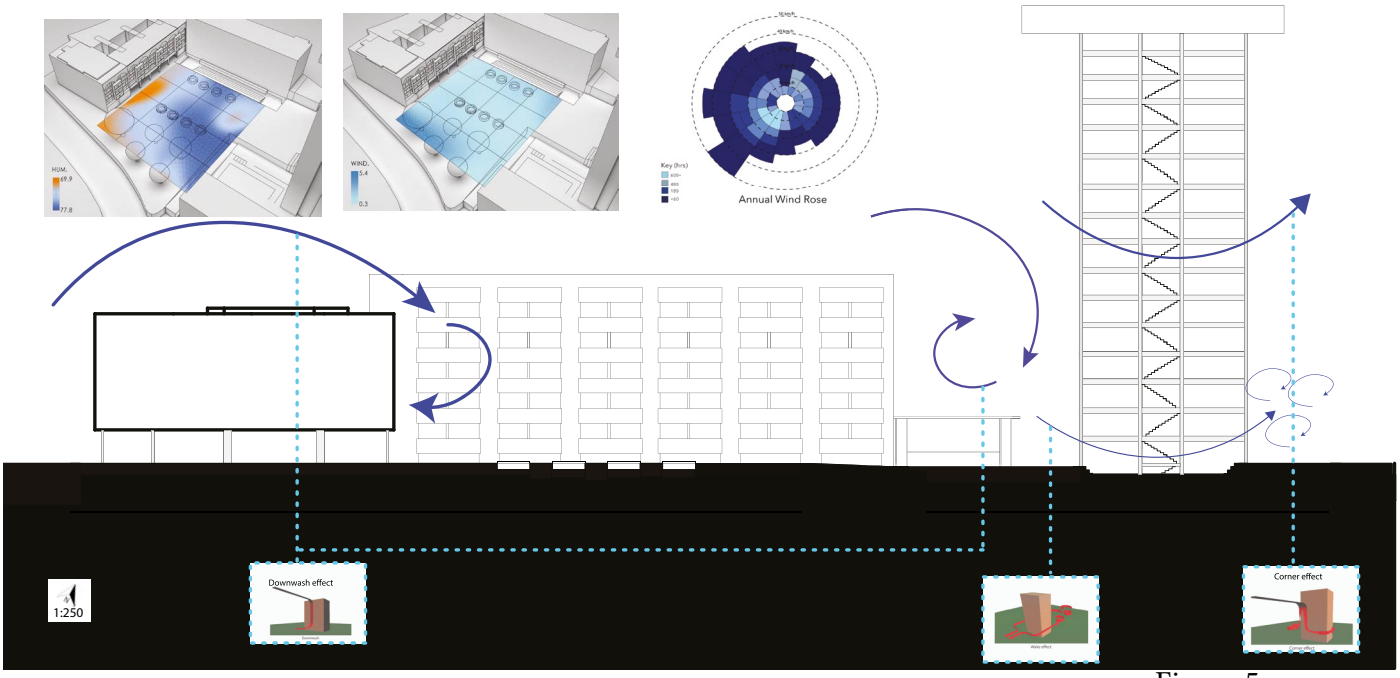



Figure 5

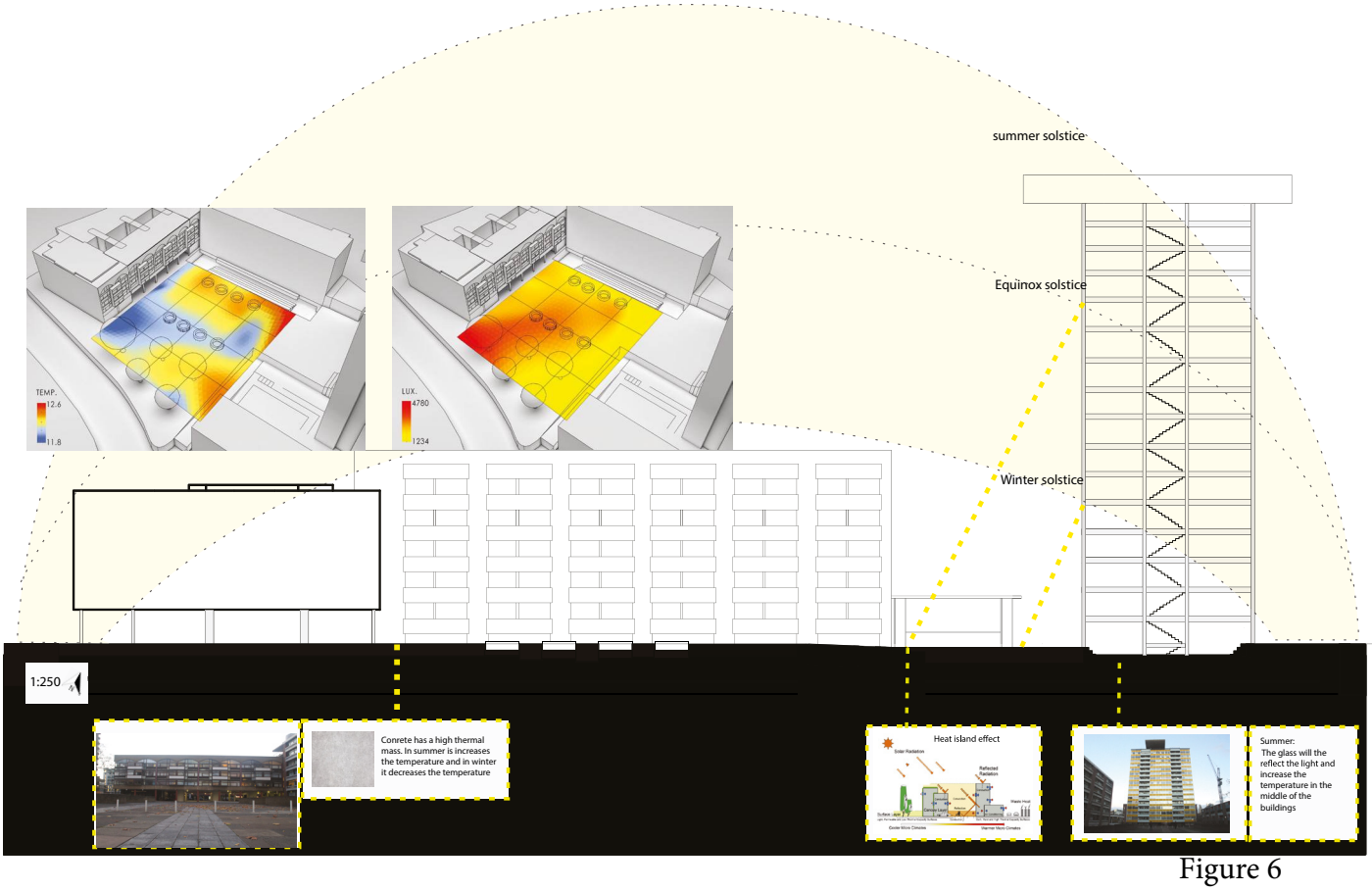



Figure 6

# Site analysis

Before
- Risk assessment
- Equipment and instrumentation check

Preparation
- Review lecture notes and literature
- Confirm the site plan
- Familiarise with procedure of experiments
- Plan: Date, time, location,equipment, instruments

Conducting
- Go through site monitoring before start
- Get ready for observing and recording results
- Carry out fieldwork and record all actions
- Review actions according to plan
- Check results and write down additional observations

Writing and report
- Process and reflect on results, and observations
- Add any additional desktop analysis (Microsoft, rhino ect)
- Consult  guidance on report writing and technical reports
- Write and illustrate following conclusions.

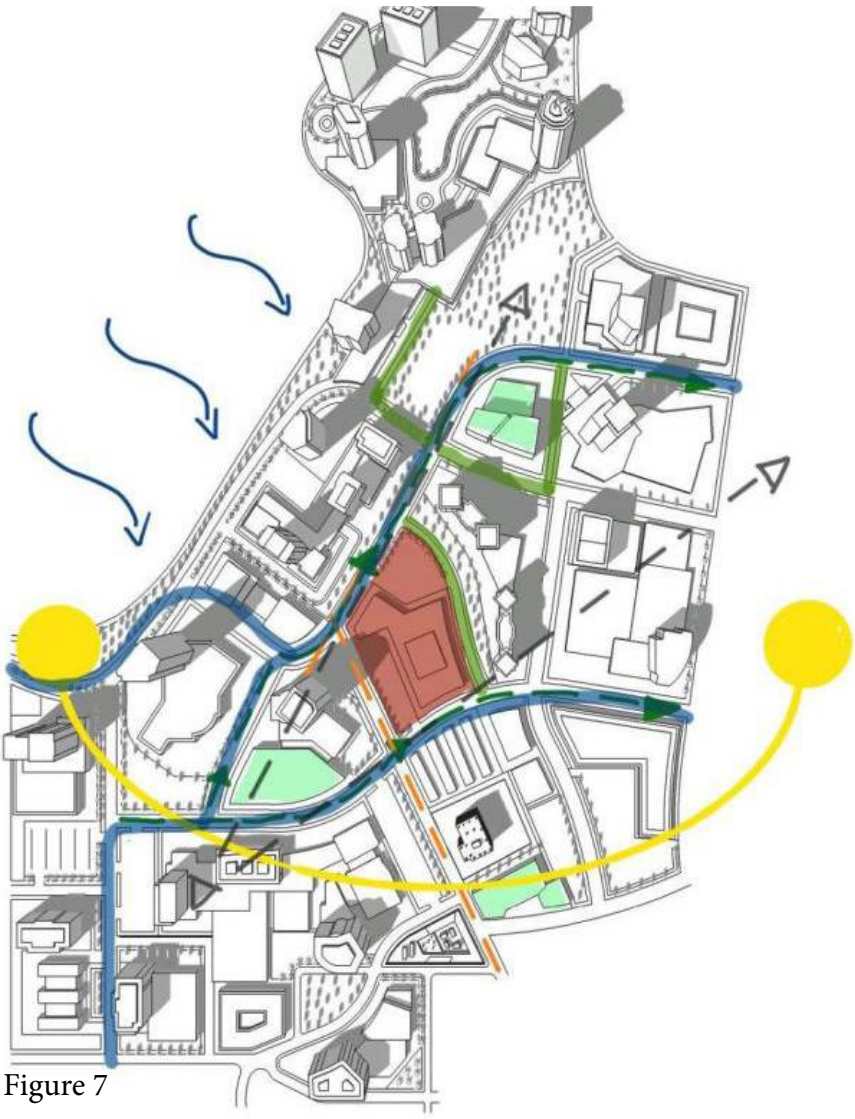


Figure 7

Key elements to analyse

- Sun

-North/South Facing
-Sun angle/azimuth
-Light
-Shadows

- Wind

-Temperature
-Direction

- Pollution

-Noise
-Air pollutants

# Design relation

This analysis allowed me to undersatnd how to conduct a microclmiate analysis with the various instruments. Allowing me to test various varibales that effect desing the core elements to enviromental design

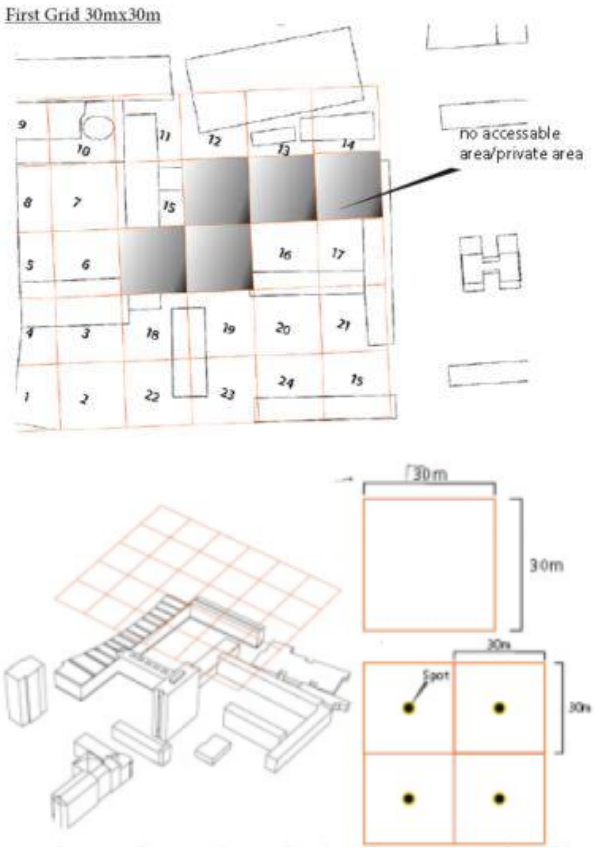



Figure 8

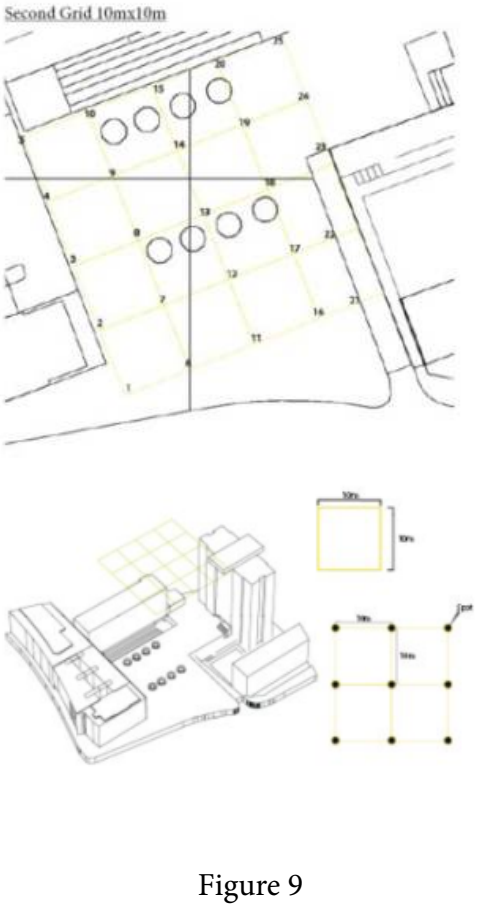






Figure 9

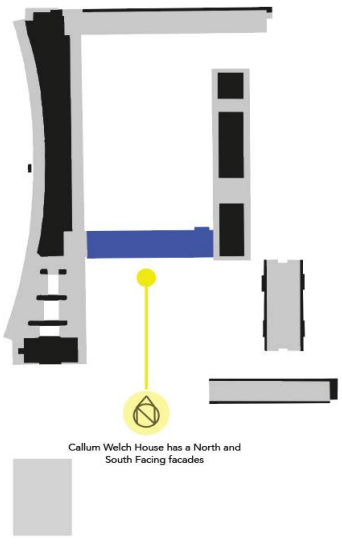



Figure 10

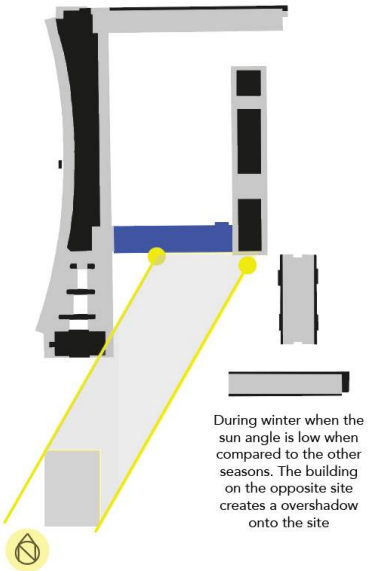



Figure 11

# Human comfort

A-Z of Architecture looking at human health and human comfort

A
- Aato Alvar- Paimio sanatorium, Finland 1932- designed to maximsie sunlight into rooms
- Adaptive comfort- boundary in which we feel comfortable
- Air quality mapping- Aries Mateus, Franciso
- Antimicrobial door hand- made from silver

B
- Breathe industires- devices to clean air
- Jason Burges- street electronic wallpapers, creates various enviroments
- Building envelope

C
- Chin Mel (1991)-bioredmedation and hyperaccumator plans- looking to improve air quality

D
- Dymaxion house 'Whichita prototype (1946)

I

E
- Evidence based design(EBD)- evidence to prove building good or bad

F
- Fanger Ole- pioneered the theory on human comfort
- Fujimato, Sou. Inspiration from shapes

G
- Graduation towers: water dropped down towers, similar to walking near ocean, Sloe berries grown inside

H
- Healing with light, natural light produced by artficail lighting.

I
- IFPRI/Cedric price- lung for midtown manhattan

K
- Killingbeck sanatorium

L
- Llewelyn- Davies, Richard,. duffle coat planning, multi startergy building.

M
- Manuel Gea Gonzalez- Prosdune elegant: titianium addative to matabolize pollution
- Mosquito train- train to fix mosquito problem in Zandria.

P
- Photon shower used by Delta express to reduce jet lag using blue light
- Pianano, Renzo

S
- Solar geometry- Altitude and Azimith
- Smog free tower
- Steiner Rudolph
- Sound mapping: sound mirros, geometry to relfect sound, barriers.

T
- Thermal imagary- visulaize heat

W
- Wind mapping



Figure 11

# Design relation

The A-Z of human comfort allowed me to find examples and architects that take human comfort into consideration when designing their buildings.

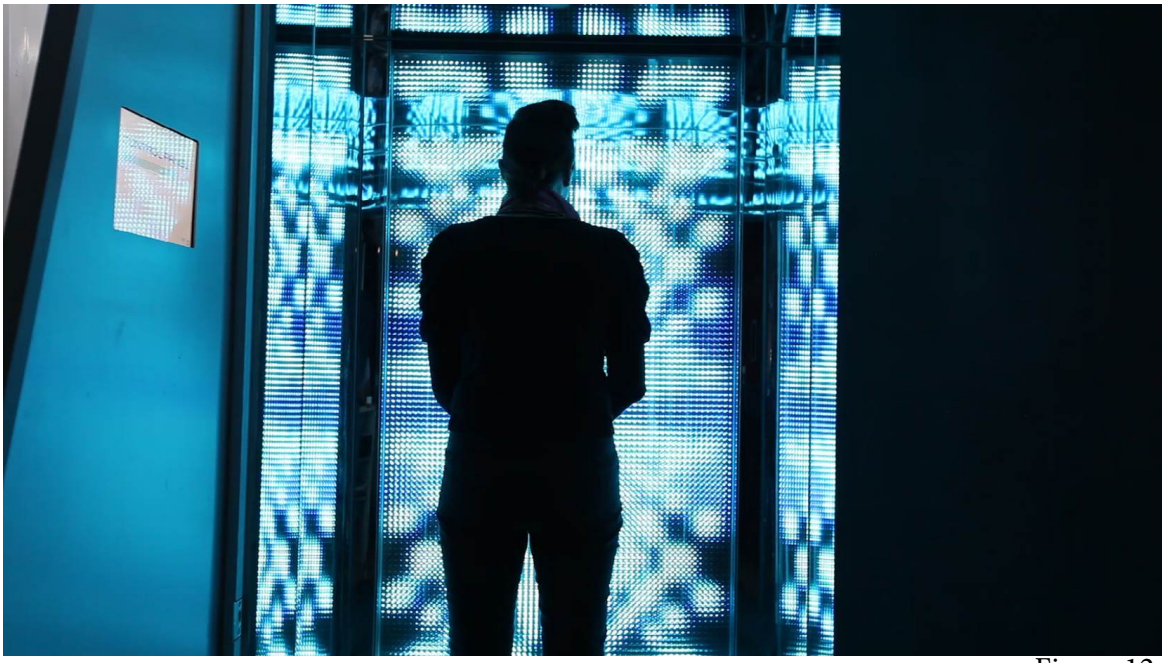


Figure 12

Using a photon shower in places that receive little light will be effective as it provides light as well as leaving people feeling refreshed and energised

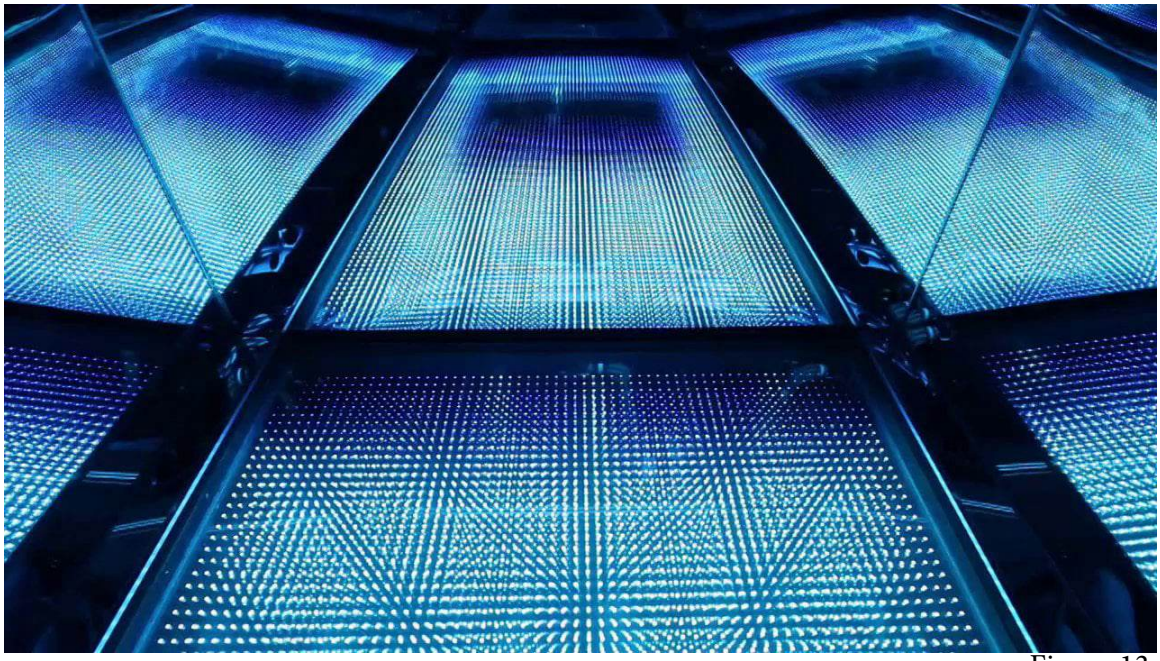


Figure 13

# Building analysis- Fieldwork

- Building analysis

Why do we analyze buildings?

- identity and  find solutions to problems in buildings
- response to our needs
- informed decision making
- improvement in future design quality

Fieldwork methods

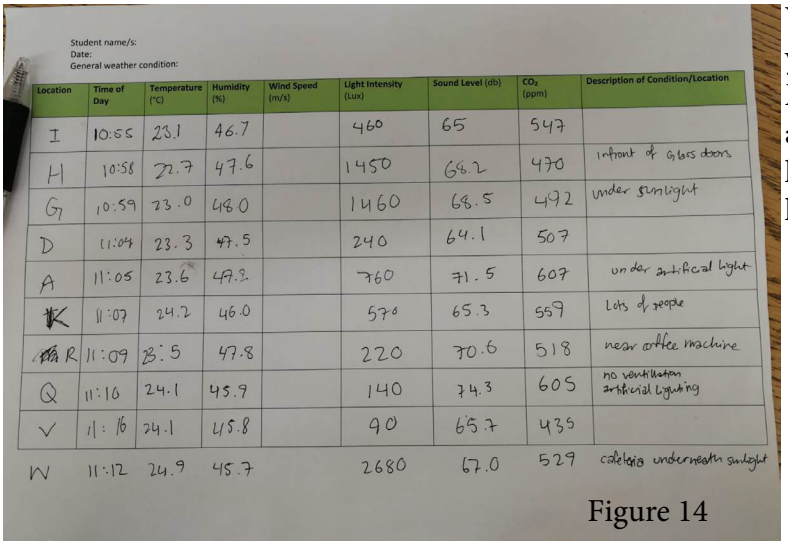
- measurements and monitoring } objective
- energy assessments } objective

- occupants surveys and interviews } subjective

Fieldwork can be undertaken anywhere

- outdoor, indoor, unbuilt

Post Occupancy Evaluation ( POE ) implies the undertaking of fieldwork on a building after occupation

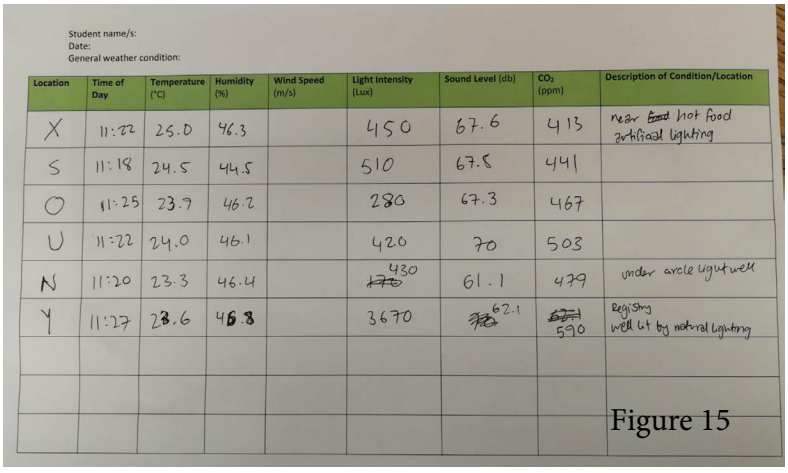


Student name/s:
Date:
General weather condition:

| Location | Time of Day | Temperature (°C) | Humidity (%) | Wind Speed (m/s) | Light Intensity (Lux) | Sound Level (db) | $CO_2$ (ppm) | Description of Condition/Location |
|---|---|---|---|---|---|---|---|---|
| I | 10:55 | 23.1 | 46.7 | | 460 | 65 | 547 | |
| H | 10:58 | 22.7 | 47.6 | | 1450 | 66.2 | 470 | infront of Gyms doors |
| G | 10:59 | 23.0 | 46.0 | | 1460 | 68.5 | 492 | under sunlight |
| D | 11:04 | 23.3 | 47.5 | | 240 | 64.1 | 507 | |
| A | 11:05 | 23.6 | 47.2 | | 760 | 71.5 | 607 | under artifical light |
| K | 11:07 | 24.2 | 46.0 | | 570 | 65.3 | 559 | Lots of people |
| R | 11:09 | 23.5 | 47.8 | | 220 | 70.6 | 518 | near coffee machine |
| Q | 11:10 | 24.1 | 45.9 | | 140 | 74.3 | 605 | no ventilation artificial lighting |
| V | 11:16 | 24.1 | 45.8 | | 90 | 65.7 | 435 | |
| W | 11:12 | 24.9 | 45.7 | | 2680 | 67.0 | 529 | cafeteria underneath sunlight |

Figure 14

Workshop where we did a site analysis for :University Of Westminster Marylebone. The aim was to find area that uses unnecessary artificial lighting or a spot that needs more lighting.



Student name/s:
Date:
General weather condition:

| Location | Time of Day | Temperature (°C) | Humidity (%) | Wind Speed (m/s) | Light Intensity (Lux) | Sound Level (db) | $CO_2$ (ppm) | Description of Condition/Location |
|---|---|---|---|---|---|---|---|---|
| X | 11:22 | 25.0 | 46.3 | | 450 | 67.6 | 413 | near food hot food artificial lighting |
| S | 11:18 | 24.5 | 44.5 | | 510 | 67.8 | 441 | |
| O | 11:25 | 23.9 | 46.2 | | 280 | 67.3 | 467 | |
| U | 11:22 | 24.0 | 46.1 | | 420 | 70 | 503 | |
| N | 11:20 | 23.3 | 46.4 | | ~~430~~ | 61.1 | 479 | under circle light well |
| Y | 11:27 | 23.6 | 46.8 | | 3670 | ~~62.1~~ | ~~590~~ | Registry well lit by natural lighting |

Figure 15

# Passive heating

Passive solar design refers to the use of the sun's energy for the heating and cooling of living spaces. In this approach, the building itself or some element of it takes advantage of natural energy characteristics in materials and air created by exposure to the sun. Passive systems are simple, have few moving parts, and require minimal maintenance and require no mechanical systems.

-Using the suns solar energy with no energy being used to draw that energy in

The five elements of passive solar gain is
- Control
-Aperature
-Distribuiton
-Absorber

-Thermal mass

## Passvive heating startergies

Single side ventilation
- Air comes in from same side
- Better air flow, openings top and bottom of window
- Ventilation dump hoovers (louvers)
- Bottom and top hoovers ( louvers )

Cross ventilation
- Least number of air changes
- Shallow room, one window each side of room

Stack ventilation
- Full building
- Greater height between inlet and outlet

Design
Single sided
- Single opening

Single sided ventilation
- Multiple openings

Well distributed
Vertically and horizontally

Size and position openings
-inlet area vs outlet area

DESIGN

- chain activators
- pampers and grills
- trickle ventilators
- bottom hung windows

Manual calculations

QV = ….x n x v x ^T
N = (p x QVR x …) /v

n = number of air changes (ac/h)
v = volume (m3)
^T = temp diff between in and out (k)
p = number of people
QVR = required ventilation rate ( 1s/p) or (m3/9)

OVR = QVA
QVA = a x v

A = opening area ( area x openable %) m2
V = air velocity ( m/s )
QVA = achieved ventilation ratio ( m3/s )

Manual calculations

Heat balance equation

Convection gain / loss

QV = 1300 x V x ^T

1300 =
V =
^T
QV = p x v x c x ^T

P = density of air (kg/m3

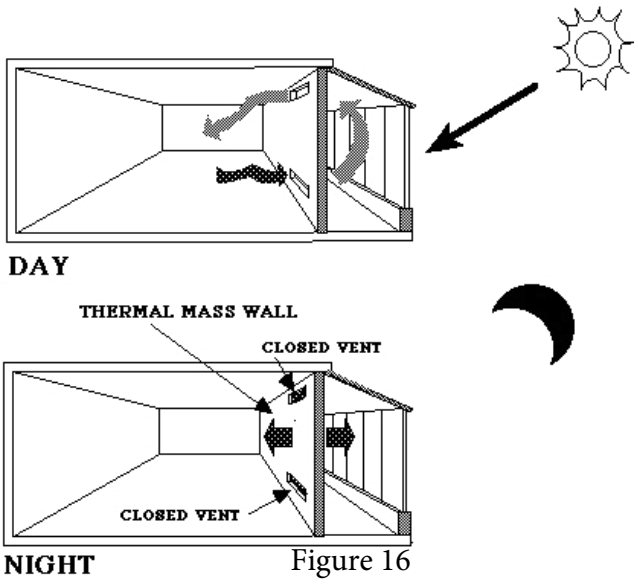



Figure 16

# Design relation

This lecture allowed me to use passive heating techniques in my desing, useful when you do not have electricity or when you have to reduce the power comsumption in a building, it is also a key element in enviromental design. I used passive heating in my design to bring in as much light and heat as possible to heat. Especailly in winter when the low sun angle allows for the suns energy to pass through the entire space

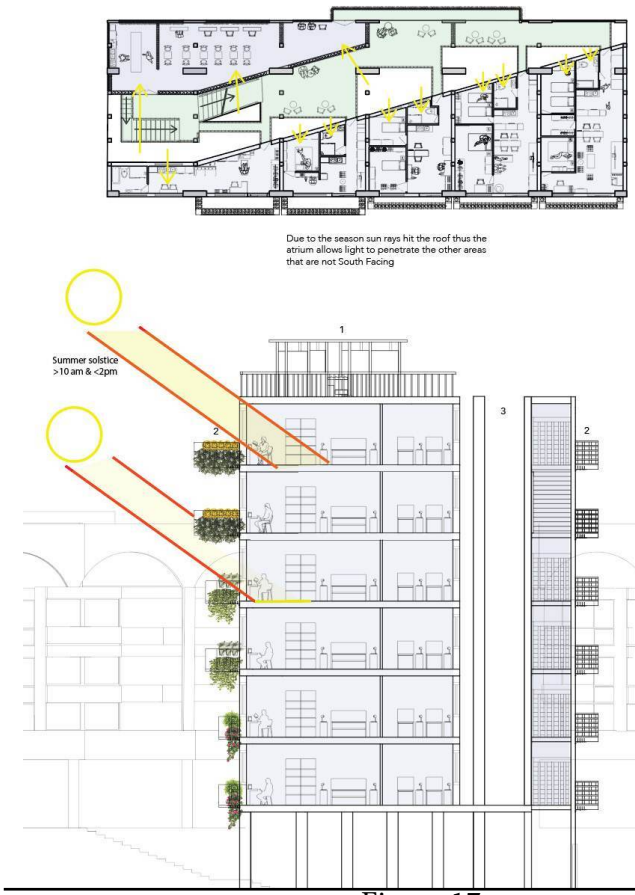



Figure 17

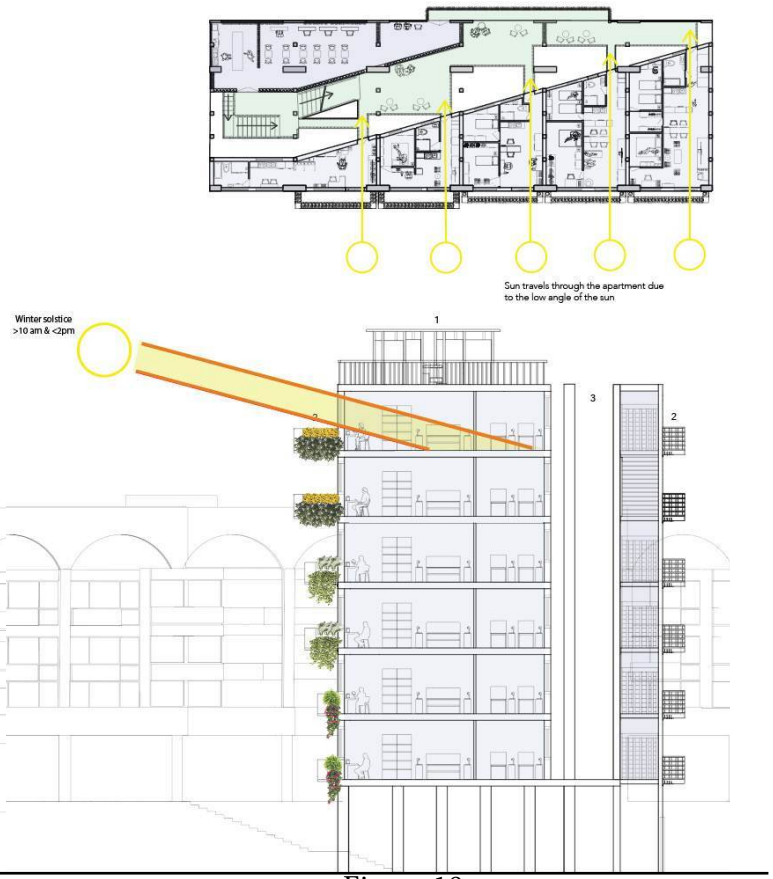



Figure 18

# Indoor mean temperature calculator

Mean Indoor calculator

Infiltration: is the process by which water on the ground surface enters the soil. It is commonly used in both hydrology and soil sciences. The infiltration capacity is defined as the maximum rate of infiltration.

- If it's high no additional ventilation needed

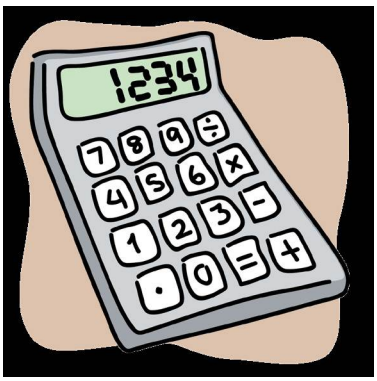


Figure 19

Ventilation: the provision of fresh air to a room/building,

Uncontrolled ventilation:

- The older the building more air loss

Extra ventilation:

- Fresh air required
- Cooling effect -remove heat

Predicted comfort temperature is the range of acceptable temperature and for human comfort
Example – if you change something what do you expect

Use the manual calculator to test differences such as:

1. Change glazing
2. Change conditions ( represent different days )

Keep in mind that this is a prediction is it not completely accurate. Use it ti find ranges (min – max) and mean average

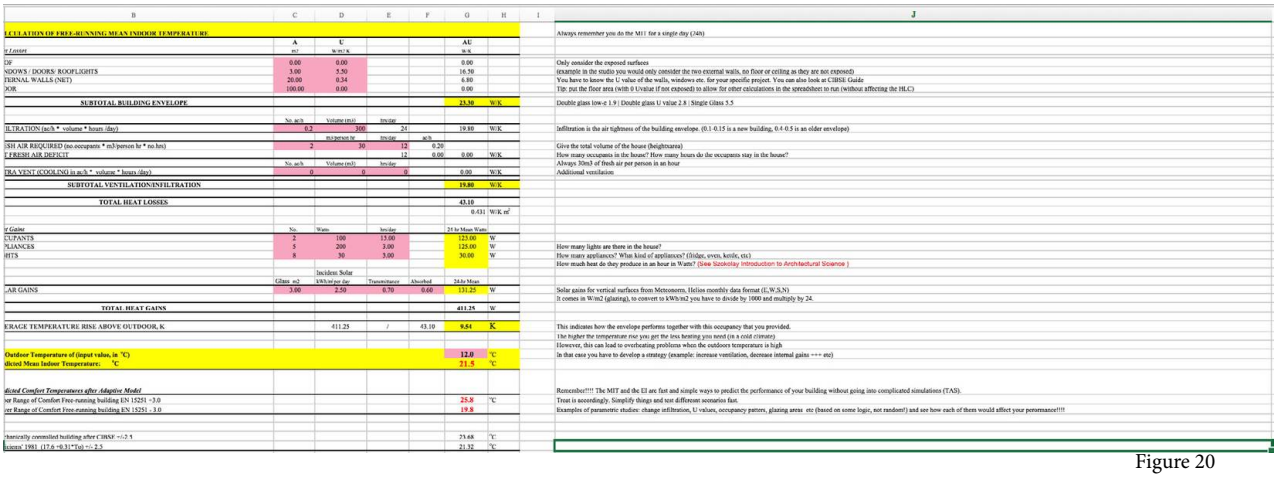
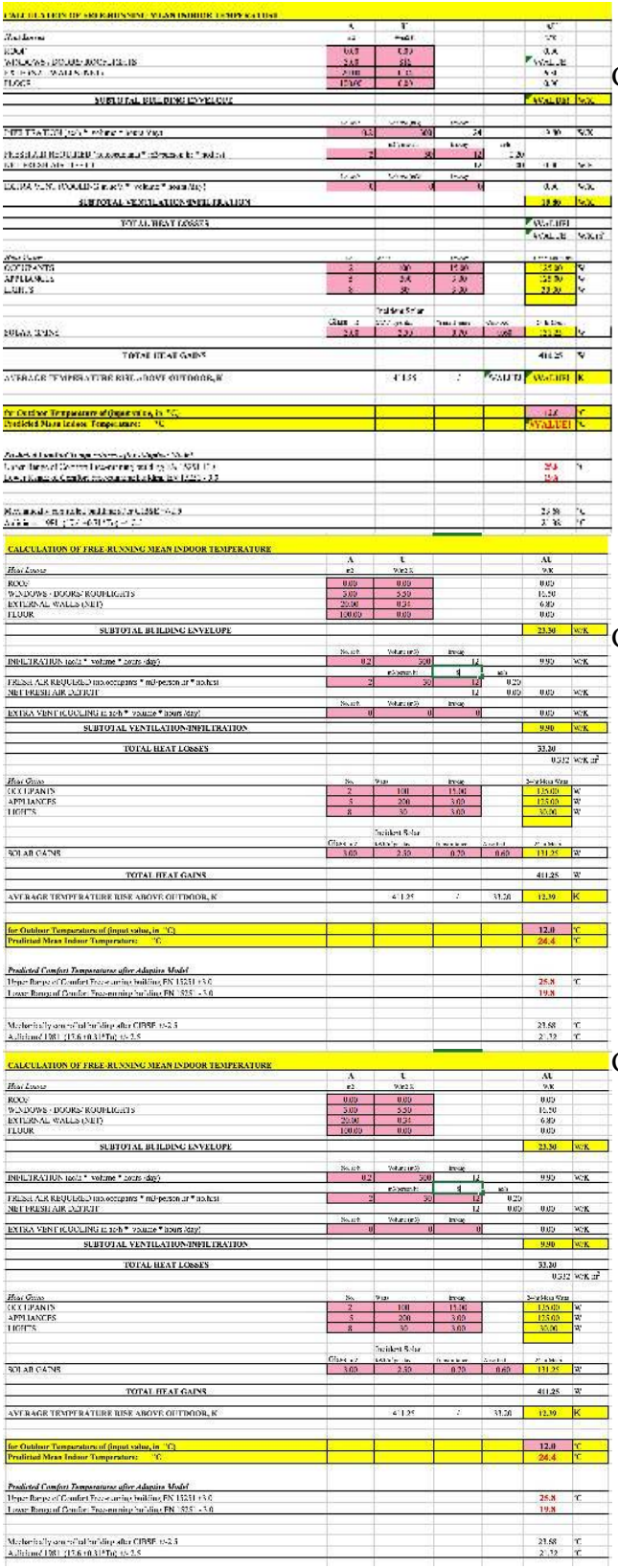


Figure 20

# Design relation

This tool allowed me to calculate the temperature in my space and find the best way to design my various areas to enure the correct temperature would be maintained.



Change double glazing

Change infiltration

Change solar gains

Figure 21

# Natural ventilation

Natural Ventilation: is the process of supplying air to and removing air from an indoor space without using mechanical systems. It refers to the flow of external air to an indoor space as a result of pressure differences arising from natural forces.

Importance
- CO2 > 1000ppm relates to inadequate air quality

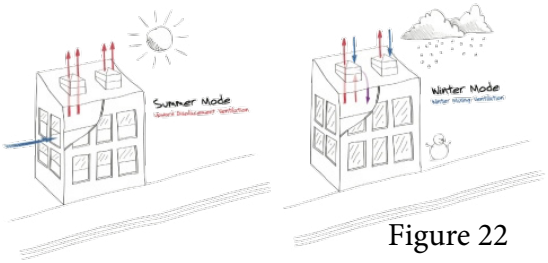


Figure 22

Natural ventilation philosophy
Summer
- Maximize ventilation for ventilation and cooling

Winter
Unoccupied
- Ensure no pollutants present
- Minimize background ventilation
- one area with both the inlet and oulet to ensure temperature is maintained

Individual rooms unoccupied
- “lights off policy “

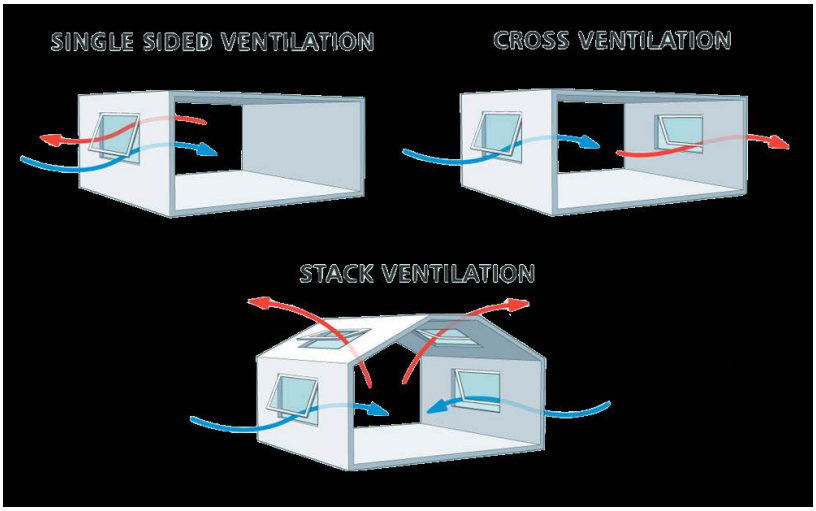
Occupied /Living/ Bedrooms
- Ventilate occupied rooms to meet the number of occupants present
- Expect ( except ) in the coldest of weather allow night temp to drift

Wet rooms
- Isolate from living rooms
- Ventilate to keep RH < 70%

VENTILATION STRATERGIES

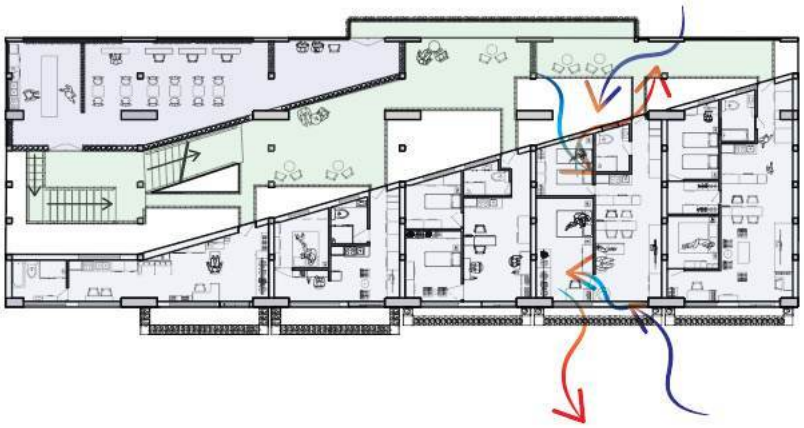
Natural
- Cross flow wind
- Wind tower
- Stack Flue
- Stack

Mixed mode ventilation
- Fan assisted stack
- Top down ventilation
- Buried pipes



# Design relation

This lecture allowed me to understand how natural ventilation techniques work and I was able to use them when designing my windows and ventilation shafts. I added a atrium to bring in air through the lower levels ventilating the inner side of the building while using single sided ventilation to cool the outer side. In winter the windows close to the atrium can be used to air the apartment while maintaing the air temperature



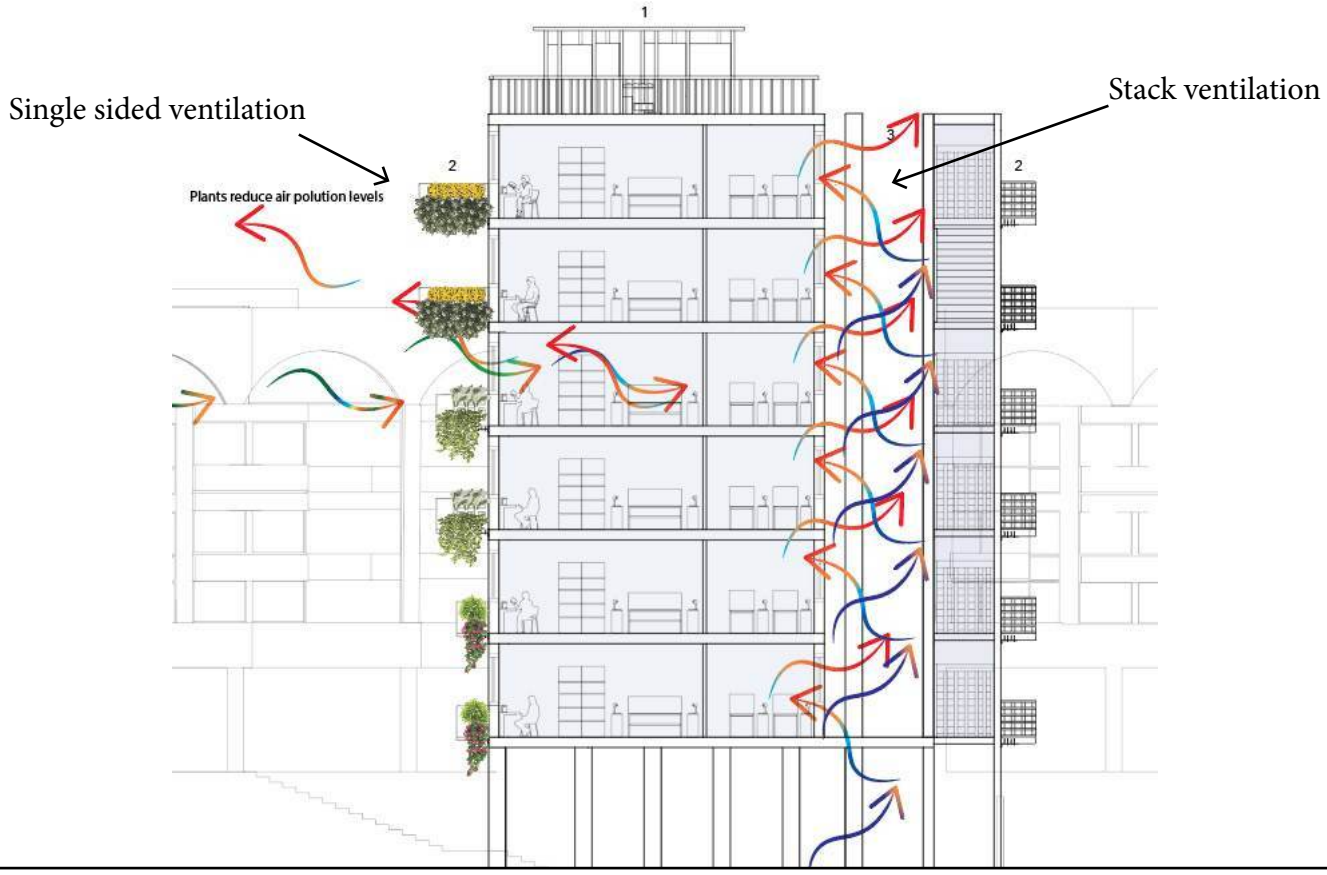
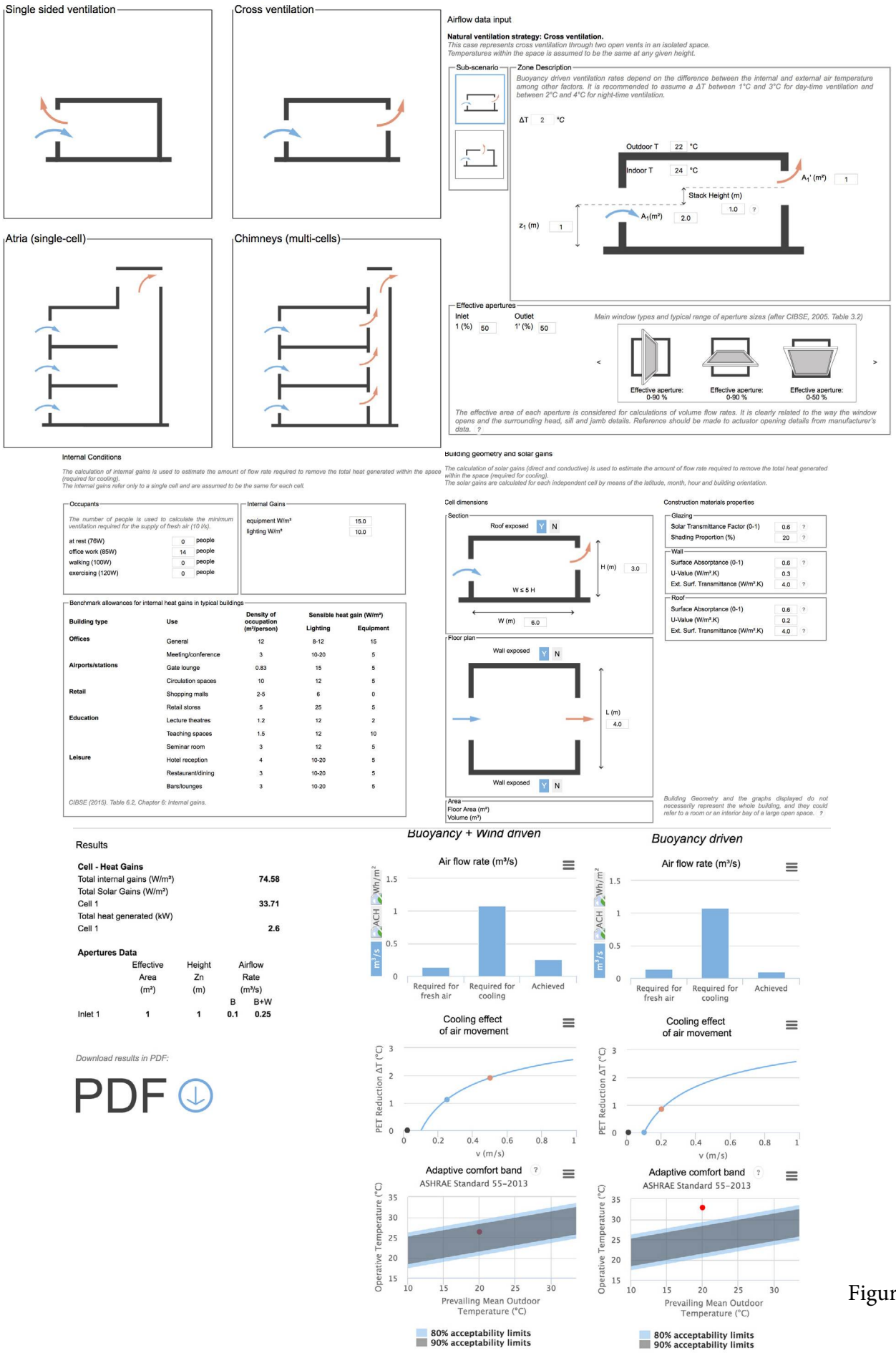



Figure 24

# Optivent tool

We were introduced to a webiste where we could calculate human comfort in a room through natural ventilation
http://naturalcooling.co.uk/OV21/optivent/optivent.php




Figures 25

# Design relation

The opvient tool allowed me to test my ventilation techniques to ensure that my design is within the accepted human comfort range. We did a excerise in class to test and learn how to use the tool. The results were as follows

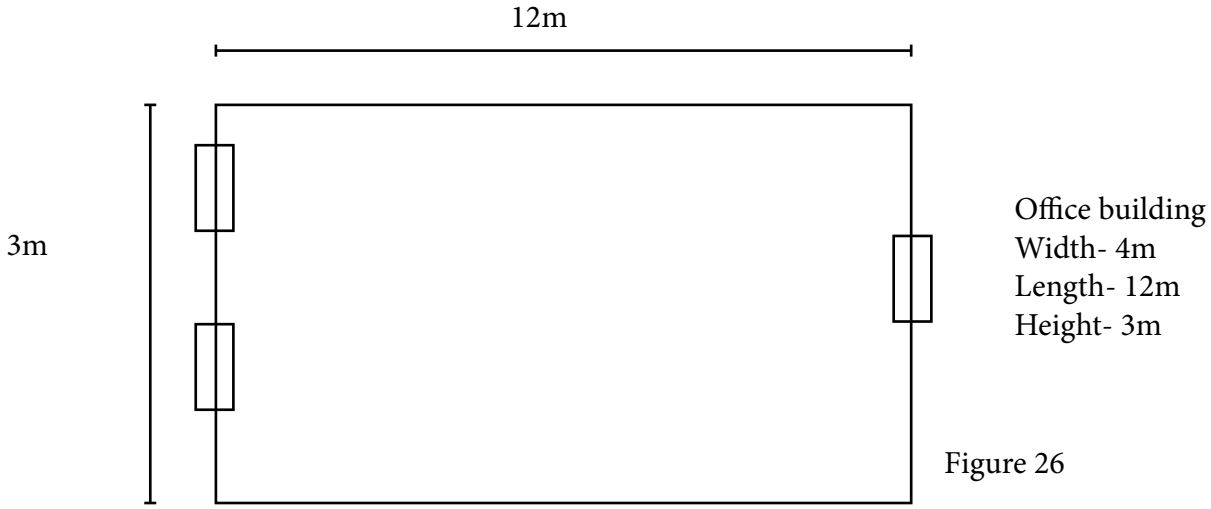



Figure 26

Maximum number of people allowed ins 15 people. During summer solstice specfcially July, where there is a outdoor temperature of 22 degrees celcius and a indoor temperature of 24 degrees celcius

The window on the left must be 2m(2) and 5m(2) on the right

# Natural cooling

The use of ventilation or natural heat sinks for heat dissipation from indoor spaces. Natural cooling can be separated into five categories different categories: ventilation, night flushing, radiative cooling, evaporative cooling, and earth coupling.

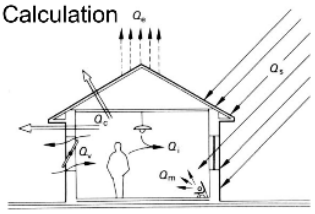
Different Passive Cooling techniques can be applied in different climatic contexts.Their success is determined by a realistic approach and clear understanding of the project brief as well as the minimization of the cooling loads.Passive cooling techniques can be successfully integrated into contemporary building construction provided the appropriate typology is chosen. Climatic applicability, performance analysis and post occupancy evaluation are crucial for successful and resilient design.

Why Natural Cooling?
- Energy crisis (crude oil price: from 19 to 90$/barrel in 40yrs)
- Heat waves of 2003 and 2006 (excess deaths in EU)
- Increased use of A/C (70% sales increase in S EU + China, India)
- Peak electricity loads and risks of blackouts in summer
- Global warming potential of refrigerants (GWP HFCs)
- Impact on health (thermal shock, bacterial contamination, etc.)

Means of Thermal Control: Heat Gains Reduction
- Low occupancy density (e.g. 14m2/p for offices)
- Use energy efficient lighting (e.g. compact fluorescent lighting)
- Use energy efficient household appliances (e.g. washing machines, fridges, etc.)
- Use energy efficient equipment (e.g. LCD screens)
- Accurately plan the location of special equipment such as photocopiers, vending machines, etc.
- Adopt a control strategy or Building Management System (BMS)

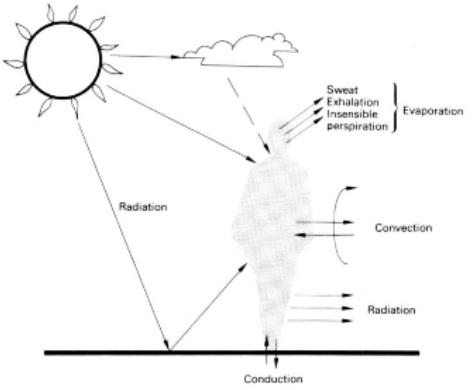
## Principles of Thermal Design

- Cooling Loads and Heat Gains Calculation
- Sources of heat gains:
- Internal Gains (Qi)
- Solar Gains (Qs)
- Conductive Gains (Qc)
- Convective or Ventilation Gains (Qv)
- Evaporative or Latent Gains (Qe)



$$Qm = Qi + Qs \pm Qc \pm Qv - Qe$$

## Principles of Thermal Design

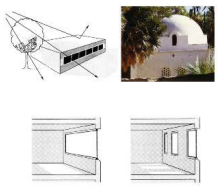



- Heat exchange in the body

## Means of Thermal Control

Solar Control by:

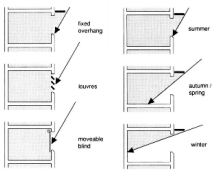
- Shading with vegetation;
- Reflective and domed roofs;
- Window design.



## Means of Thermal Control

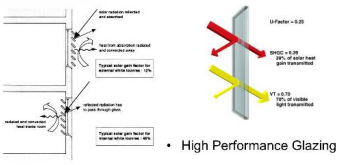
Solar devices:

- Overhang;
- Louvres;
- Blinds.




## Means of Thermal Control

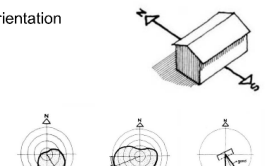
- Internal vs External Blinds



- High Performance Glazing

## Means of Thermal Control

- Orientation



*a) solar radiation in winter; b) solar radiation in summer; c) optimum orientation*

Figures 27

# Design relation

I used 1.5m windows to maximise the amount of light entering the building. A atrium is also added to cool the inner part of the building. Vegetation is also used to help provide shade in summer. Wind passes through each apartment via single sided ventilation.

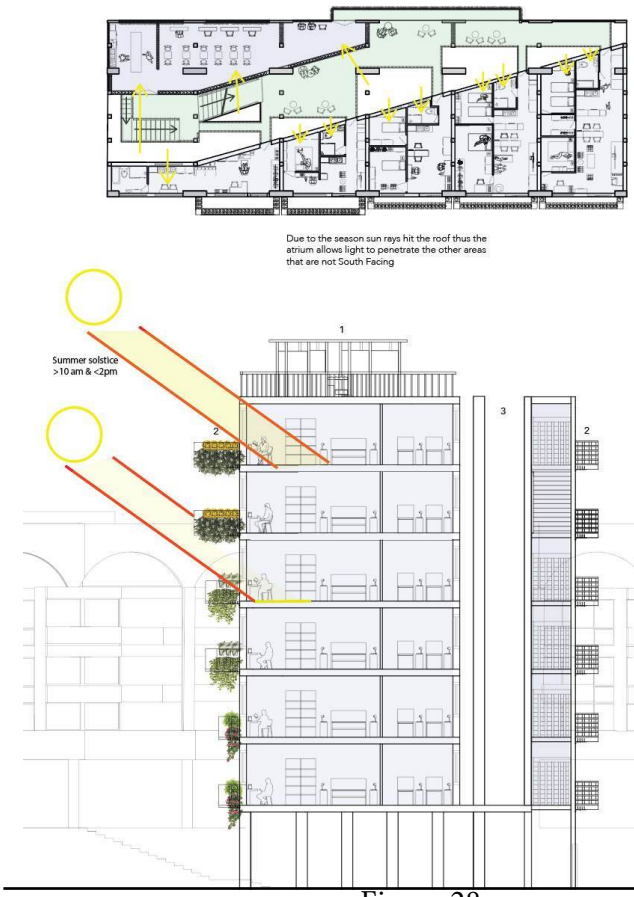



Figure 28

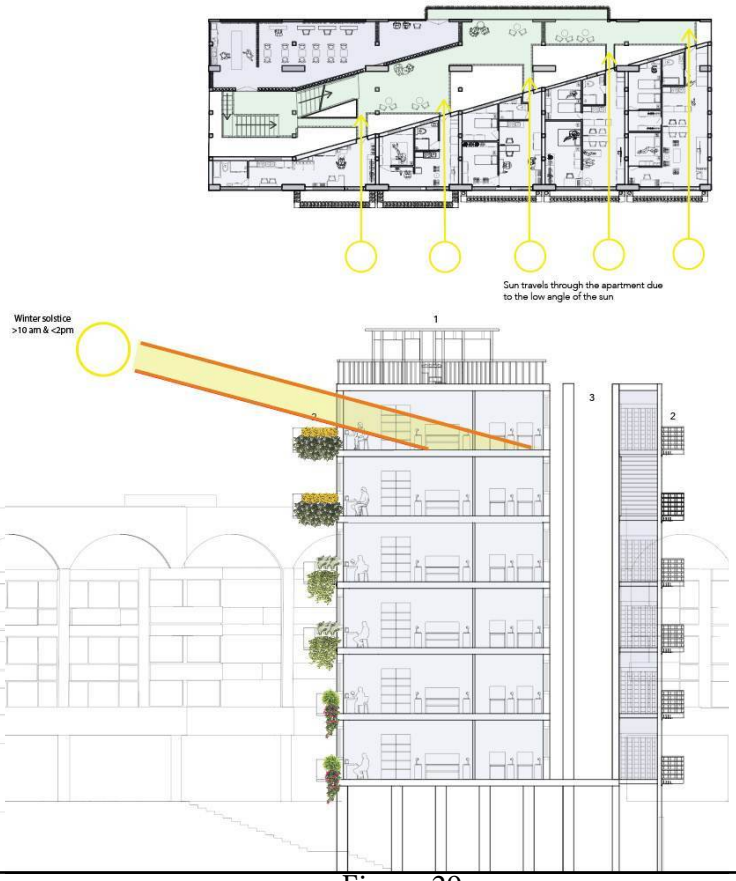



Figure 29

# Building envelope

A building envelope is the physical separator between the conditioned and unconditioned environment of a building including the resistance to air, water, heat, light, and noise transfer.

Thermally Massive elements have an inherent 'time-lag' using Thermal Mass. Cooler night air used to pre-cool the building and dispel the previous day's heat loads using Thermal Mass and Stack Effect

During occupation thermal mass assists cooling of internal environment Using Thermal Mass and Stack Effect

## Styles

Model of a typical South European Medieval Village incorporates many tried and tested thermal and ventilation strategies Using Thermal Mass and Stack Effect

The design of a traditional Moorish Courtyard, evolved over 100's of years and still in use today, to provide a comfortable environment in the fiercest heat of the day.

## Use:

Building envelopes can be used to enhance the performance or solve surrent issues.

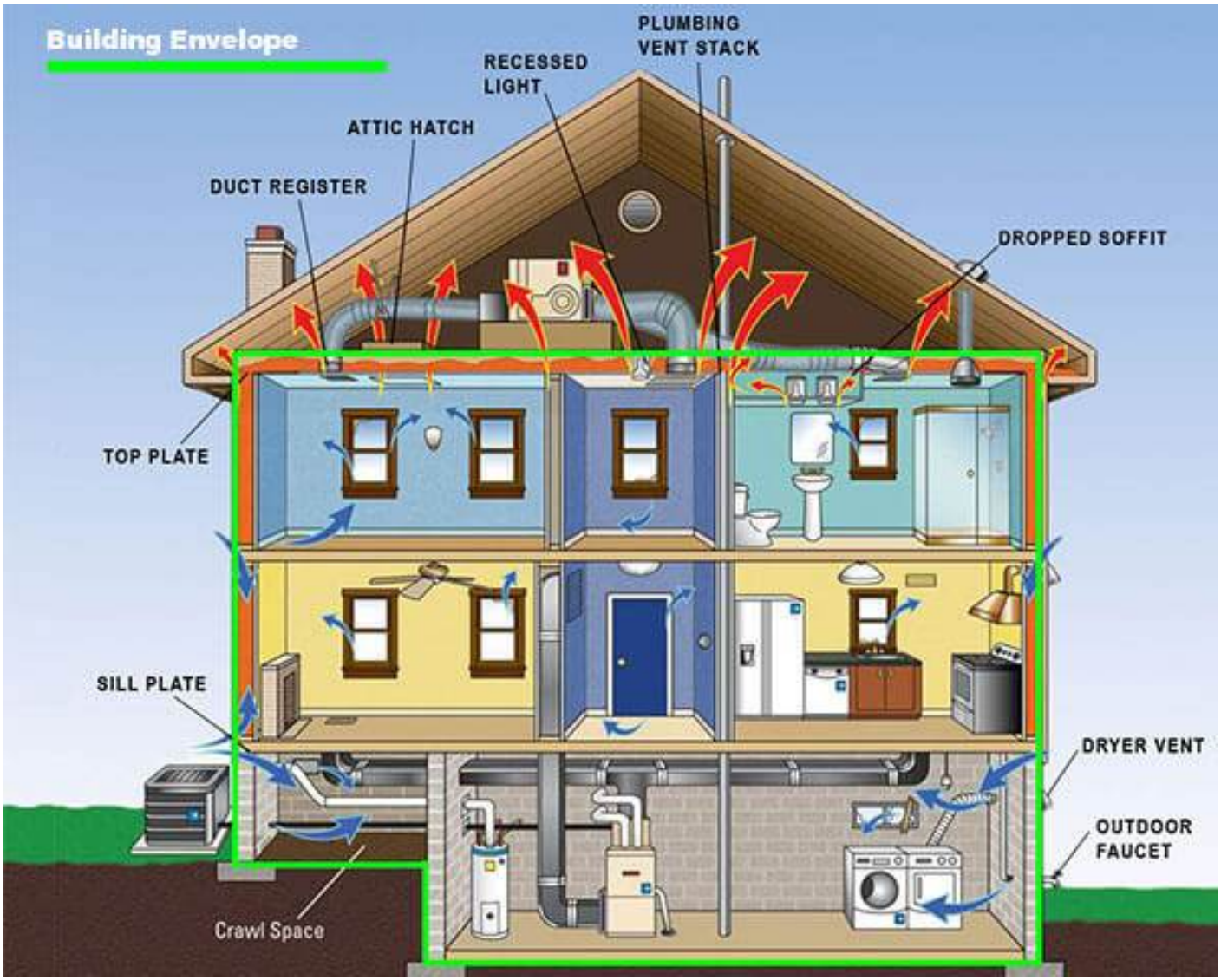



Figure 30

# Designing for solar control

Solar control has been used to enhance the effectiveness of design since a early age. Le Corbusier is a key figure when it comes to solar control. Using different textures and sun angles he manages to manipulate light to ensure his designs utilize light in the most effective and sufficient way possible.

We visited Le Torette which was built by Le Corbusier. As you can see he uses light tunnels and in the rooms a smooth texture is added to ensure light does not reflect onto the table where you can do work.



Figure 33

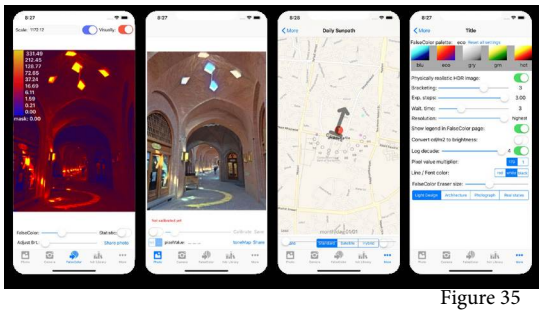
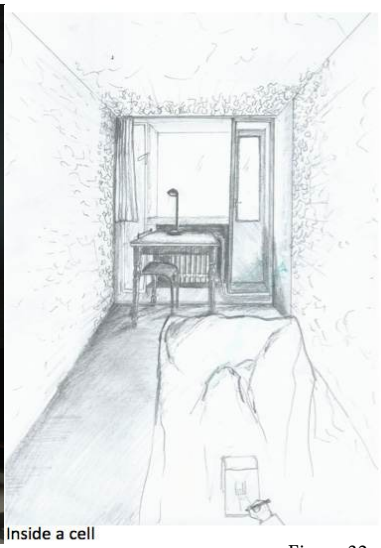


Figure 35



Figure 31



Inside a cell

Figure 32

Aftab was introduced to use and it is a luminance mobile application that can be used for analysing light.

By creating a real HDR image that can be downloaded or uploaded via iTunes to or from a computer, this app is suited for four different users that are as follows:

1- Lighting Design
2- Architecture
3- Photographing




Figure 34

# Thermal Comfort

•Thermal Comfort is the condition of mind that expresses satisfaction with the thermal environment and is assessed by subjective evaluation (ANSI/ASHRAE Standard 55).
•The human body will generate excess heat into the environment, so the body can continue to operate.
•Maintaining this standard of thermal comfort for occupants of buildings or other enclosures is one of the important goals of HVAC (heating, ventilation, and air conditioning) design engineers.
•Most people will feel comfortable at room temperature, colloquially a range of temperatures around 20 to 22 °C but this may vary greatly between individuals and depending on factors such as activity level, clothing, and humidity.
•Thermal neutrality is maintained when the heat generated by human metabolism is allowed to dissipate, thus maintaining thermal equilibrium with the surroundings.
•The main factors that influence thermal comfort are those that determine heat gain and loss, namely metabolic rate, clothing insulation, air temperature, mean radiant temperature, air speed and relative humidity

How to analyse buildings:
- Measurements & Monitoring -(Objective)- Prediction through
- Energy Assessments -(Subjective)- User thermal comfort survey

Adaptable Opportunities
- Openable windows - Adjustable Blinds - Desk fan or locally controlled ceiling fan - Local heating/cooling controls - Workstation/furniture flexibility - Shallow plan (minimising distance from windows) - Cellular rooms (reduces mutual disturbance) - Good views - Appropriate surface finishes - Daylight and task lighting backups

Empirical and adaptive methods Empirical methods:
Based on lab tests mostly focusing on the physiological responses of the human body subject to controlled environmental conditions
PMV (Predicted mean vote)/PPD (Predicted percentage of dissatisfaction) An index that predicts the mean value of the cotes of a huge group of persons on the 7-point thermal sensation scale
OT (Oparity temperature/ Resultant temperature: DBT+Mean Radiance/2) Uniform temperature of an imaginary enclosure in which an occupant would exchange the same amount of radiant heat. (Maximum frequency allowed in 2.5%)

Adaptive Methods: Developed from field studies of people in their daily life to capture the variations in comfort expectations

- Humphreys (1976)

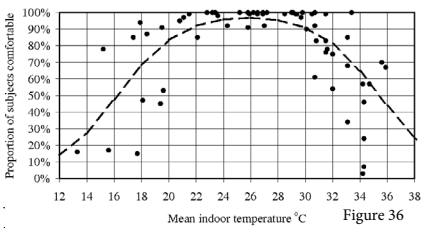


Figure 36

Formula : T om = (T omax+T omin)/2

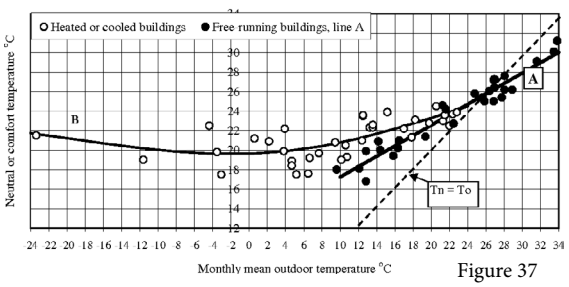


Figure 37

How to undertake a Post Occupancy Evaluation (Post Occupancy Evaluation)? Occupant Perspective
Occupant Comfort Surveys: Questionnaire Design
1.Clarify objectives
2.Choose occupants Sample *Make sure occupants have consistent experience of building
3.Type of Question Presence or absence (tick boxes/multiple choice)

Ashrae Scale
- Keep questionnaire as short as possible
- Make it personal (collect in person)
- Keep record of where subjects are located(plan)
- How many subjects to survey

| Value | Sensation |
|---|---|
| +3 | Hot |
| +2 | Warm |
| +1 | Slightly warm |
| 0 | Neutral |
| -1 | Slightly cool |
| -2 | Cool |
| -3 | Cold |

Figure 38

# Thermal comfort workshop

**Thermal comfort Workshop (comfort voting)**

| Subject ID | Personal information | | Time | Personal Factor | | Subjective Vote | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Gender | Age group | | Clo value | Activity level | Thermal Sensation: How are you feeling at the moment? | | | | | | | Preference: What kind of improvement you prefer at the moment? | | | | |
| | (Male/Female) | 15-19 \| 20-24 \| 25-30 \| 30-34 \| 35-40 \| above | | (Between 0 - 3.5) | Seated \| Standing relaxed \| Standing working \| Reclining \| Walking \| Running \| Other | Cold | Cool | Slightly cool | Neutral | Slightly warm | Warm | Hot | Much Warmer | Slightly Warmer | No change | Slightly cooler | Much cooler |
| | | | | | | -3 | -2 | -1 | 0 | 1 | 2 | 3 | -2 | -1 | 0 | 1 | 2 |
| w16104965 | Male | 22 | 10:30 | 1 | Seated | | | | 0 | | | | | | 0 | | |
| | | | 10:40 | 1 | Seated | | | | 0 | | | | | | 0 | | |
| | | | 10:50 | 1 | Seated | | | | 0 | | | | | | 0 | | |
| | | | 11:00 | 1 | Seated | | | | 0 | | | | | | 0 | | |
| | | | 11:10 | 1 | Seated | | | | 0 | | | | | | 0 | | |
| | | | 11:20 | 1 | seated | | | | 1 | | | | | | 0 | | |
| | | | 11:30 | 1 | Walking | | | | 1 | | | | | | 0 | | |
| | | | 11:40 | 1 | seated | | | | -1 | | | | | | -1 | | |
| | | | 11:50 | 2 | Walking | | | | 0 | | | | | | -1 | | |

Figure 39

This workshop gave us a hands on approach teaching us about thermal comfort. Doing experiments in the classroom where we would take off our clothes and change the window openings. All testing how thermal comfort can change depending on the situation.

While looking at thermal comfort we were introduced to smart home technologies and management systems all surrounding thermal comfort, through smart home control devices.

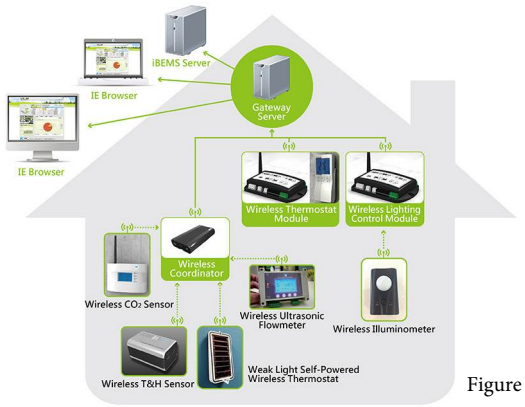
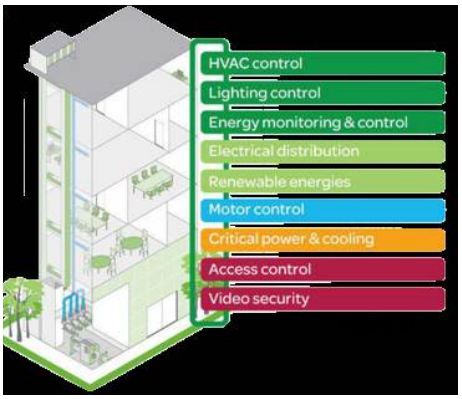



Figure 40

Building energy management systems (BEMS).
Smart energy meters
Internet of things (IoT)

These systems all make the process of regulating energy in buildings easy and efficient. They give real time data and allow trained professionals to monitor the data instantly.




An example of a thermal comfort survey taken from:

https://www.researchgate.net/publication/258140202_A_Field_Study_on_Thermal_Comfort_of_Occupants_and_Acceptable_Neutral_Temperature_at_the_National_Museum_in_Malaysia/figures?lo=1

**THERMAL ENVIRONMENT SURVEY**

*This survey is part of a study to evaluate the current thermal conditions of the selected building. We appreciate your feedback in this evaluation. Please tick at the square box where applicable.*

1. Gender: Male ☐ Female ☐
2. Age :
3. Occupant location:
4. Occupant's Clothing

*Please refer to the attached Table 1. Place a check mark next to the articles of clothing that you are currently wearing as you fill out this sheet. If you are wearing articles of clothing not listed in the table, please enter them into the space provided below.*

Clothing:

5. Occupant Activity Level *(Tick the one that is most suitable)*

| | | | |
|---|---|---|---|
| Seated quite/writing, 1.0met | ☐ | Walking about, 1.7met | ☐ |
| Typing, 1.1met | ☐ | Lifting/packing, 2.1met | ☐ |
| Standing relaxed/Filing(seated), 1.2met | ☐ | Light machine work, 2.2met | ☐ |
| Filing(standing), 1.4met | ☐ | Heavy machine work, 4.0met | ☐ |

6. How would you describe your typical level of thermal comfort?

| | | | |
|---|---|---|---|
| +3 Hot | ☐ | -1 Slightly Cool | ☐ |
| +2 Warm | ☐ | -2 Cool | ☐ |
| +1 Slightly Warm | ☐ | -3 Cold | ☐ |
| 0 Neutral | ☐ | | |

***Table 1***
***Clothing Ensembles***

| Description | |
|---|---|
| Trousers, short-sleeve shirt | |
| Trousers, long-sleeve shirt | |
| Trousers, long-sleeve shirt plus suit jacket | |
| Trousers, long-sleeve shirt plus suit jacket, vest, T-shirt | |
| Trousers, long-sleeve shirt plus long sleeve sweater, T-shirt | |
| Trousers, long-sleeve shirt plus long sleeve sweater, T-shirt plus suit jacket, long underwear bottoms | |
| Knee-length skirt, short sleeve-shirt (sandals) | |
| Knee-length skirt, long sleeve-shirt, full slip | |
| Knee-length skirt, long sleeve-shirt, half slip, long-sleeve sweater | |
| Angle-length skirt, long-sleeve shirt, suit jacket | |
| Walking-shorts, short-sleeve shirt | |
| Long-sleeve coveralls, T-shirt | |
| Overalls, long-sleeve shirt, T-shirt | |
| Insulated coveralls, long-sleeve thermal underwear tops and bottoms | |
| Sweat pants, sweat shirt | |

Figure 42

# Energy generation by Chris Coonick

Climate, energy and architecture

20th Feb

Chris Coonick working at B.R.E: introdcution
- Use of renewable energy
- Latest innovations used in the industry
- Degree in electrical energy
- Masters in renewable energy
- Works in property development
- Helps with eden project
- Always looking  to improve projects through innovation
- Green building hub in cornwell
- UK exceeded renewable energy capacity surpasses fossil fuels for first time
- Big increase compared to five years ago
- Renewable electricity
- Solar photovoltaics
- Wind turbines

Technology types
-heat
-hybrid
-electricity

Considerations. Planning  permissions: if renewable energy is used, a building's permission is granted with ease and less time.

- building regulations
-system integrations
-structural capabilities
-system performance
-maintainability
-suitability
-cost



Figure 43

## Solar Panels

- Every hour 100 000 solar moduels are installed acorss the world
- PV effect found in 1920s; sunlight placed on a pm junction with glazing ontop and absorbing material, sun shines on the silicone cell and is absorebed
- Types;
- monocrystalline cell: 22%, 25% more expensive,
- polycrystalline cell: waste materials from electronic industry, 16%,
- Amorphous thin flims- reduce manufacturing costs on SP, reduce weight, reduce weight, technology is sprayed on, not as efficient,
- OPV cells; organic solar panels that can be placed on windows
- The pitch is important; 35 degrees is best performance in UK, minimum of 10 degrees

## BIPV

- Different shapes and colours
- Energy through art
- Sun roofs and shading
- Coda building by onyx solar
- Glass brick walls with Solar PV inside

## Components

- Generating panels
- DC/AC converter need access

## Wind turbines

- UK was best wind resource in Europe
- 7 727 onshore+1934= 21 000 000 m homes powered
- smaller scale means less wind is covered and less energy is kept
- bent trees mean good location for wind turbines
- curved blades lead to more energy being captured
- helical turbines
- higher heights lead to more KW being captured
- better in open rural spaces away from urban areas, buildings create wind vortexes that offset the wind that the turbines need
- Noabal wind map

## Hydro power

- One of oldest forms of electric generation
- Water wheels, for gridnig grain
- Archimedes screw, crossflow, Kaplan, francis, turgo, pelton
- Reserved storage
- Hydropower storage plant
- Hover dam
- Pumped storage
- Electricity is cheaper at night as less people use it
- Offshore hydro: UK has some of the best tidal basins
- Tides are predictable

## Heat pumps

- Refrigerator in reverse
- Ground source, air source,
- Works well with passive house
- Can be placed in foundations

## Ground source

- Boer hole
-

## Solar thermal

- Anything with high solar load works well with solar thermal
- Evacuated tube collectors

Hy-brid technologies

PV-T
- Niche market
- Solar collector with pv panel
- Less efficient when they are hot
- Stagnentation you need somewhere for the heat to go

Geothermal
- Deep underground thus the need for a clear understanding of the geology
- Hot core deep within earths curst
- Similar to gas turbine
- Renewable depending on gas

Biomass
- Garbage, wood, crops, alcohol fuels, landfill gas,
- Needs ceratin amounts of room, specific environment, costly to run set up but cheaper to produce and in the long run
- Has harmful effects especially when burning dry and moist wood.
- Biomass boilers need constant wood

Heat pumps
- Refrigerator in reverse
- Ground source, air source,
- Works well with passive house
- Can be placed in foundations

Ground source
- Boer hole

Solar thermal
- Anything with high solar load works well with solar thermal
- Evacuated tube collectors

Biomass
- Garbage, wood, crops, alcohol fuels, landfill gas,
- Needs ceratin amounts of room, specific environment, costly to run set up but cheaper to produce and in the long run
- Has harmful effects especially when burning dry and moist wood.
- Biomass boilers need constant wood

Hybrid technologies

PV-T
- Niche market
- Solar collector with pv panel
- Less efficient when they are hot
- Stagnentation you need somewhere for the heat to go

Geothermal
- Hot core deep within earths curst
- Similar to gas turbine
- Renewable depending on gas

Hydrogen fuel cell (personal question)

- Fuel for trucks
- Dangerous
- Good energy storage medium
- You can store left over energy in hydrogen
- Transfer hydrogen into a dense form
- Then use an Energy transfer

# Design relation

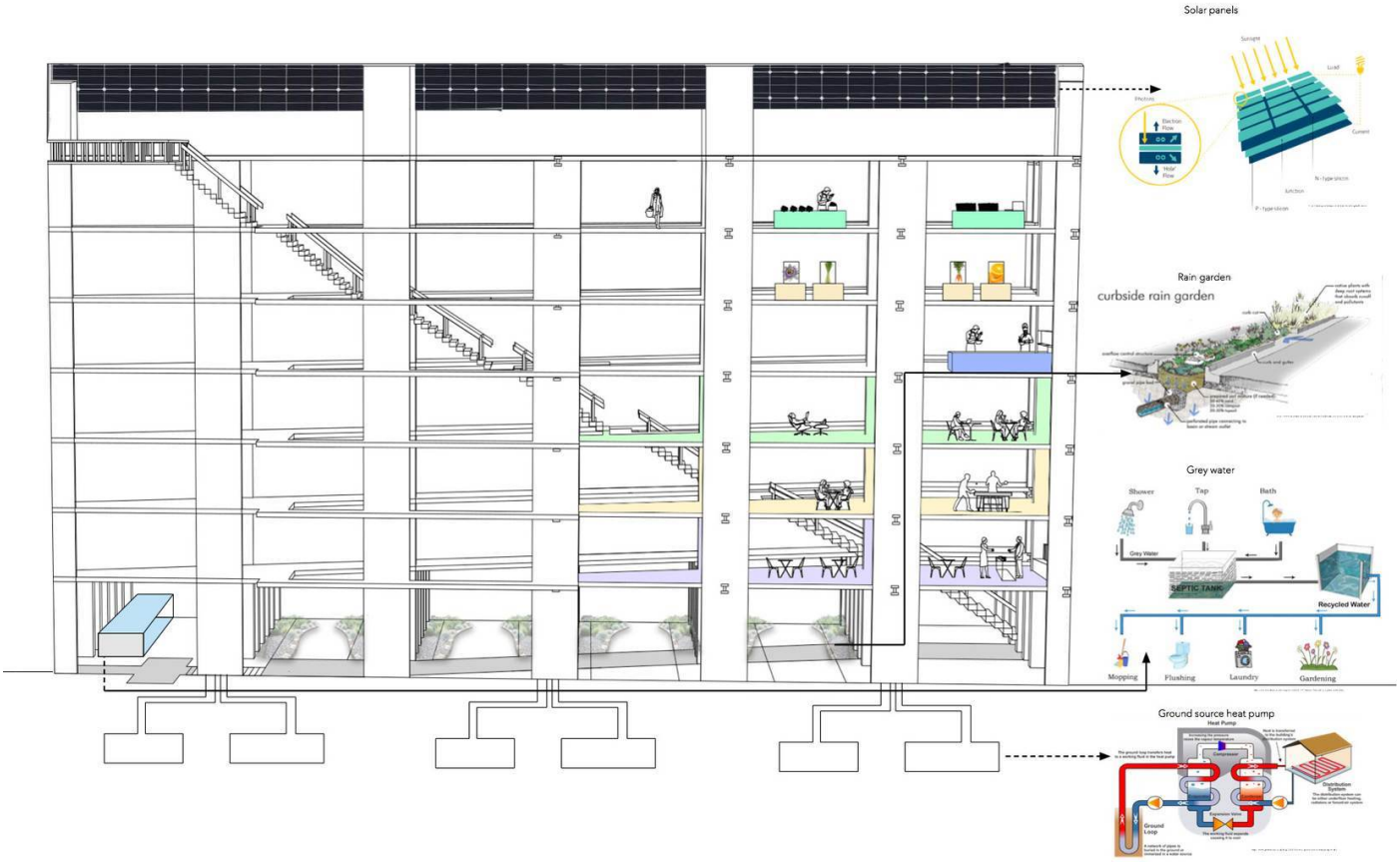



Figure 44

Systems in my desing to help me acheicve a zero-plus energy building

- Solar panels
- Rain garden
- Grey water
- Ground source heating

# Saint Gobain

The Saint Gobain tour was extremely useful. It introduced us to new innovative technologies that we can use as designers to improve our designs. The tour showed us various brands and businesses that are new to the market.

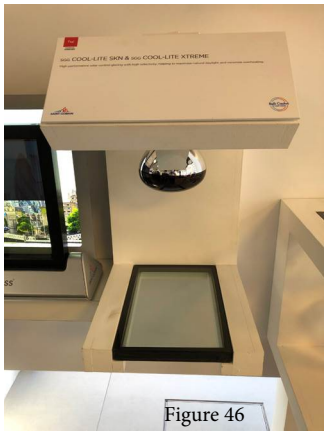
Ensign Eezi Fit (cast iron system)
- 20(DB) difference from PVC pipes
- makes flushes quieter



Figure 45

priva lite glass
- glass with an internal filter that can block the view
- opaque to transparent

SGG Planitherm total +
- Glass that does not heat up when exposed to a heat source

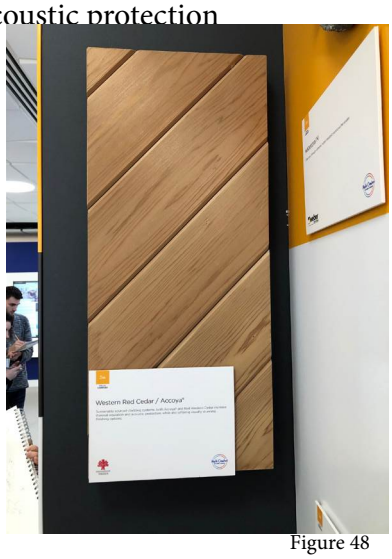

Figure 46

Sage glass
- Allows users to choose the best level of natural light year around



Figure 47

Accoya
- Sustainable cladding systems
- Increase thermal and acoustic protection
- Longevity


Figure 48

# Future build exhabition

The Future-build Events team works closely with the industry's associations, media, bellwether and new emerging suppliers to deliver what the industry needs most from its chosen events, always putting the spotlight on true innovation and excellence, and providing the stage for essential debate and knowledge share around fundamental industry issues.

The exhibition introduced us to a number of new and upcoming building technology.



Figure 49

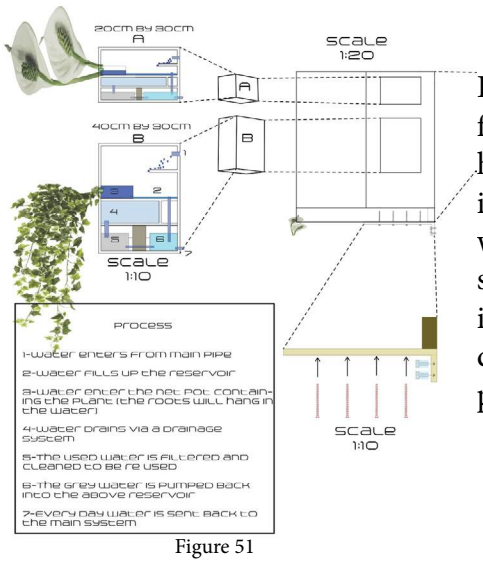


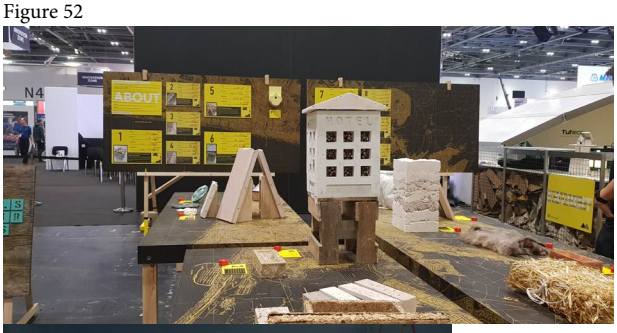
Figure 51

I am introducing an urban farm into my design so the hydroponic system was very interesting, using human waste to fertilize the crops. The system its self was extremely interesting and it helped me developed my hydroponic pods

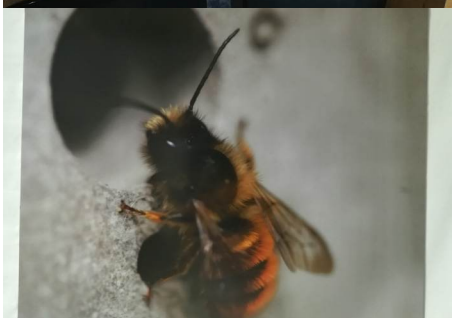


Figure 50

Figure 52





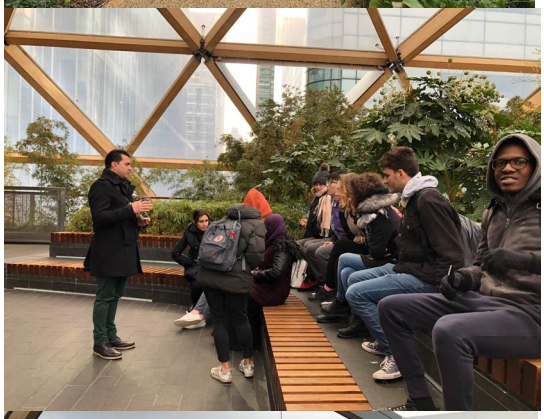
To the left you can find bug hotels. Systems used to increase the amount of bugs in cities, more specifically bees which are on the verge of becoming extinct, using these hotels in urban designs is defiantly a solution for current problems that can be used to fix future endings.

# Crossrail Place Roof Garden Canary Wharf

Figure 53











Biophilia effect: The biophilia hypothesis also called BET suggests that humans possess an innate tendency to seek connections with nature and other forms of life.

Heat island effect: An urban heat island is an urban area or metropolitan area that is significantly warmer than its surrounding rural areas due to human activities. The temperature difference usually is larger at night than during the day, and is most apparent when winds are weak

Canary docks used for shipping now its banking. History

Pollution effect on people
Primary and secondary pollution
Amount of air we breathe is more than amount of water and good needed.
10kgs per day air

Benefits of greenery

- relaxing
- Reduce heat island
- Biodiversity

Sticky hairy leaves can be used to trap pollution
Different lines of defence to battle pollution

Design relation




Figure 54

I used a number of different plants in my design to help combat all the pollutants within a city. The roof garden gave me inpiration to include vegatation into my design.

# Conservation of power and energy by DR Luisa Brotas

Paris Cop agreement 2015
- Avoid global increase in temperatures
- Below 2c above prehistoric industrial levels

16 December 2002 agreement was made

- Energy performance certificate
- Part L- conservation of power
- Part F- ventilation
-

EPC – energy performance certificate
DEC- display energy certificate
Every 10 years you have to go through re-evaluation

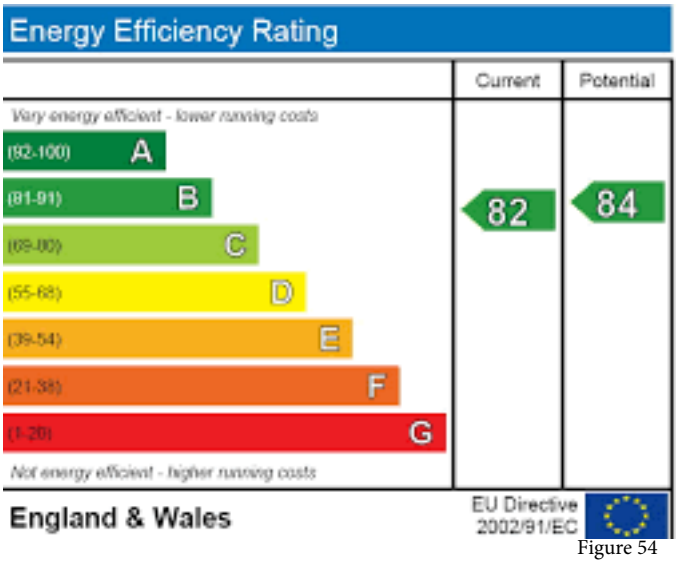


Figure 54

Zero carbon policy
- All new buildings must be nearly zero energy buildings by 2020 (nZEB)
- Reduce UKS emissions of co2 by 34% by 2020 and 80% by 2050
- High standards of fabric energy efficiency to reduce energy demand

- Heat accounts for a third of the United Kingdoms greenhouse gas emissions

Main energy use in buildings

- Space heating
- Water heating
- Cooking
- Lighting and appliances

Zero carbon homes
- Goal is to reduce 35% of carbon emissions

Greater London Authority hierarchy
- Be lean
- Be clean
- Be green

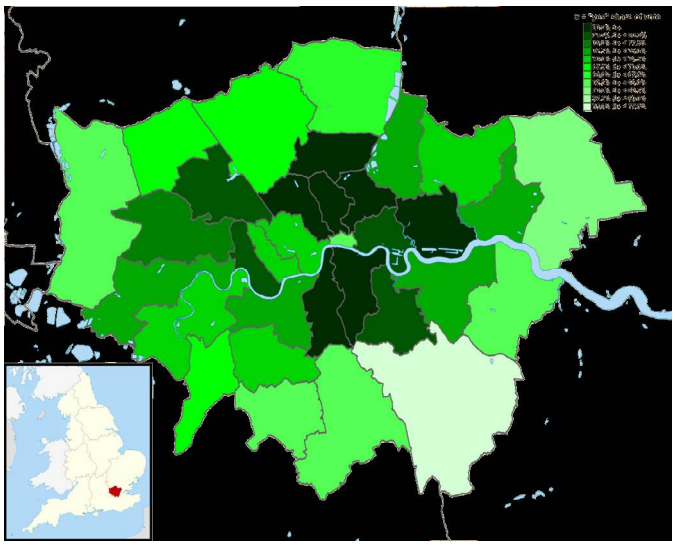
Greater London Authority - Cooling hierarchy
Managing heat risk
- Minimize internal heat generation
- Reduce amount of heat entering a building through orientation, shading, albedo, fenestration, insulation, and green roofs/walls
- Manage heat within the building through exposed internal thermal mass and high ceilings
- Provide passive ventilation
- Provide mechanical ventilation
- Provide active cooling systems

Be lean
-cavity wall properties
- properties with a loft
- solid wall properties – most potential for improvements

Be clean

Greater London Authority hierarchy
- Be lean
- Be clean
- Be green

Greater London Authority - Cooling hierarchy
Managing heat risk
-Minimize internal heat generation
-Reduce amount of heat entering a building through orientation, shading, albedo, fenestration, insulation, and green roofs/walls
-Manage heat within the building through exposed internal thermal mass and high ceilings
-Provide passive ventilation
-Provide mechanical ventilation
-Provide active cooling systems



Be lean
Be clean
Be green
- Heat pumps
- Co-efficiency of performance

Low carbon tech
- Generation of heat for space and hot water
- - high efficiency

Standard Assessment Procedure
- standard assessment procedure for energy rating
- assess energy performance

Part L1A New dwellings

Compliance steps
- achieving TER, TFEE, DER, TER, DFEE, TFEE
- limits of desing flexibility
- limiting the effects of heat gains in summer
- building performance consistant with DER
- Provisions for energy efficient operation of the dwelling

Opaque vs glazing – notional building

Reduce U Value

Ratings
SAP- E cost/m2 (1-100)
Environmental impact rating (EL): co2 ann./m2 (1-100)
Dwelling co2 emission rate (DER): C02 ann/ m2 kg/m2/year

Order of examination for performance

- Production
- Construction
- Operational
- Demolition

Factors that contribute to efficiency
- Materials
- Thermal Insulation
- Air leakage ventilation
- Efficiency
- Solar gains
- Fuel
- Energy use for space cooling
- Renewable energy
-

Types in order of enrgy used

- Detached
- Semi detached
- mid terrace
- end terrace
- enclosed terrace
- enclosed mid terrace
- enclosed end terrace

dwellings classification
- house
- bungalow
- flat
- maisonette
- park home

# Design relation

Timber vs concrete

Concrete

- Concrete- polluted from processing – bad production
- Concrete- is better for performance
- CLT- light thermal mass but air conditioning is needed, more operational consumption

CLT

- Less time for construction
- Cheaper
- Less emissions overall
- Use face change materials
- Thermal mass addition needed
- Overheating issues
- Zero- energy building is possible
- Durability
- Concrete centre
- Stop air from entering then deal with issues inside





Figure 57

# Water systems by Cath Hassell

What it is about
- Considerations for sustainable water designs
- Design with real context
- It will give you the questions you need to ask


Figure 58

Natural cycle of water

Chalk streams
-mains water
-hot water
-waste water

Water use in your building
- Sanitary
- Cooking

Solutions for sustaible water use

- Dual flush WCs
- flow regulation at taps and heads
- automatic taps
- internal and logged meters
- urinal controls or waterless urinals
- total washroom control

Carbon load on water
- 1.2kwh of energy to supply and treat 1m3 of water
- 0.7kg of co2 emissions per m3 of mains water used
- 6% of domestic and total carbon emissions from water use

Heating
- solar thermal
- roof integrated or roof pitched
- can supply 50-70% of buildings heating

Solar PV panels to heat water
- placed on the roof of a building
-

Heat recovery from grey water
- 62% of waste heat is recovered
- good potential for centralised systems in hotels

Reed beds/contrasted wetlands
- sewage treatment plant with foliage
- biodiversity into a site
- phragmites australis

Sustainable drainage systems (SUDS) 1
- attenuate, treat, and dispose of storm water to reduce pollution and local flooding
-

Sustainable drainage systems (SUDS) 2
- green roofs and rainwater harvesting
- permeable paving
- rain gardens
- tree pits

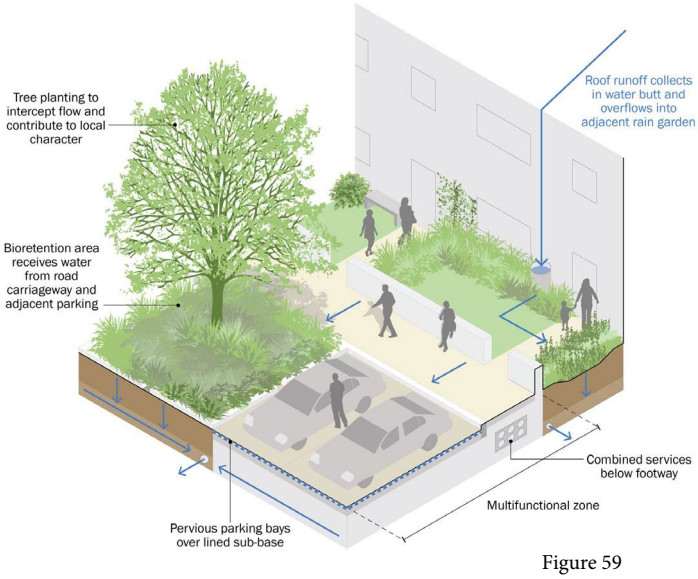


Figure 59

Fish farms of Kolkata
- closed loop system
- 30% of cities sewage provides food for fish farm in east Kolkata
- 3500 hectares
- 13 000 tonnes of fish year. 150 tonnes of vegetables a day of irrigated with the waste water
- livelihood for 10 000
- better to have it out of the city, difficult to get the planning in London

# Design relation

Cooling effect of vegetation

Green roof

- provide biodiversity
- sub strata provide the attenuation
- intensive, extensive , bio-diverse
- well suited to commercial/

Green walls

- some require lots of maintenance
- Belgrave building near Victoria

Rainwater

- fit for: WC,urinal flushing, washing machines, and grounds watering
- Rain water harvesting

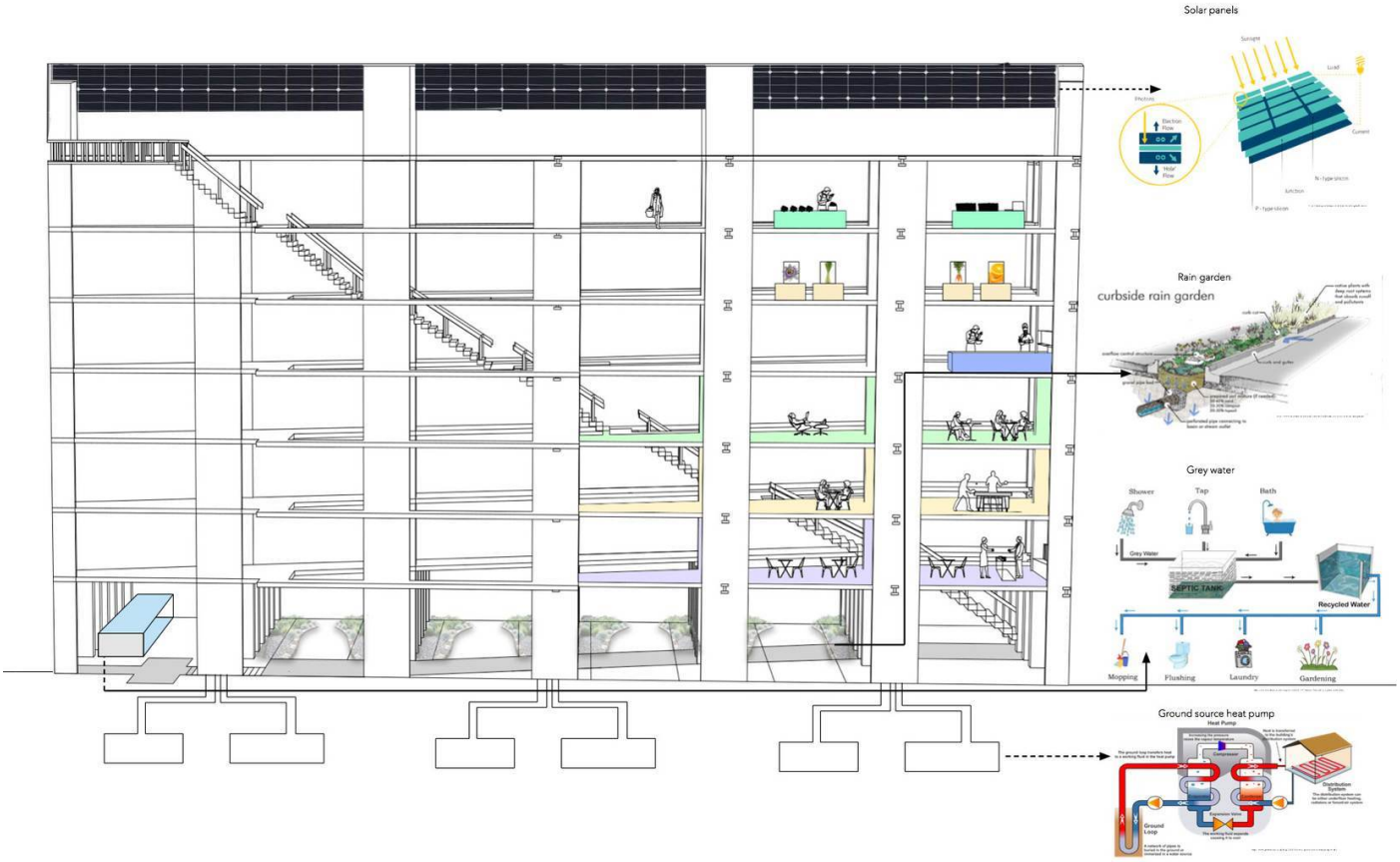



Figure 60

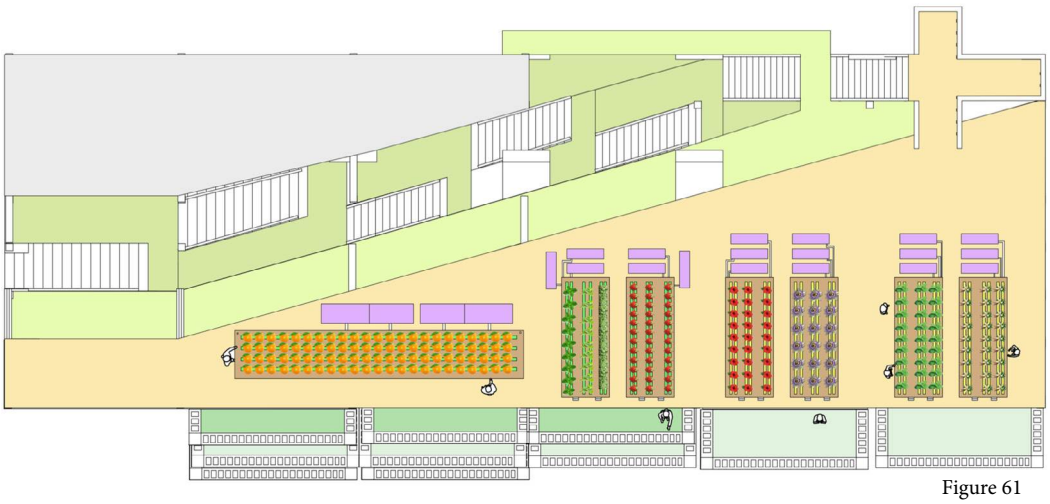


Figure 61

# The Crystal



Figure 62

Smart buildings

Breeam: BREEAM New Construction is the BREEAM standard against which the sustainability of new, non-residential buildings in the UK is assessed. Developers and their project teams use the scheme at key stages in the design and procurement process to measure, evaluate, improve and reflect the performance of their buildings.

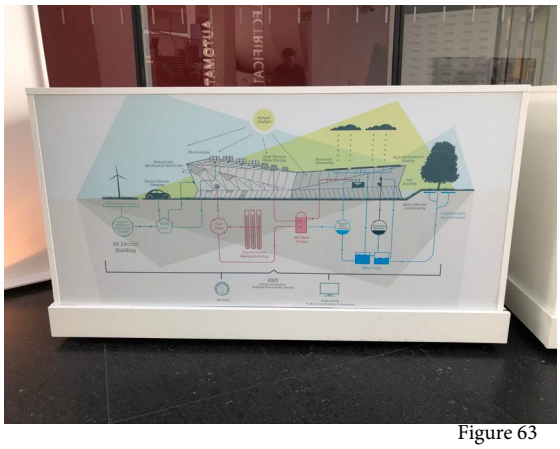


Figure 63

Rio- environmental meeting agreement for climate change. Legally binding agreement backed by the Paris agreement. Steps to a brighter future
Increasing population means increasing in pressure put on the planet leading to negative after effects

We can live longer but planning and organisation needs to be changed. Sustainable design needs to be encourage in all aspects of life. The crystal showcases different sustainable technologies.

We had to complete a game to get a A grade building rating which introduced us to different methods of sustainable design

Methods for sustainable design:
Heat pump
Ground source heating pump
Green tariff




Figure 64




Figure 65



Figure 66

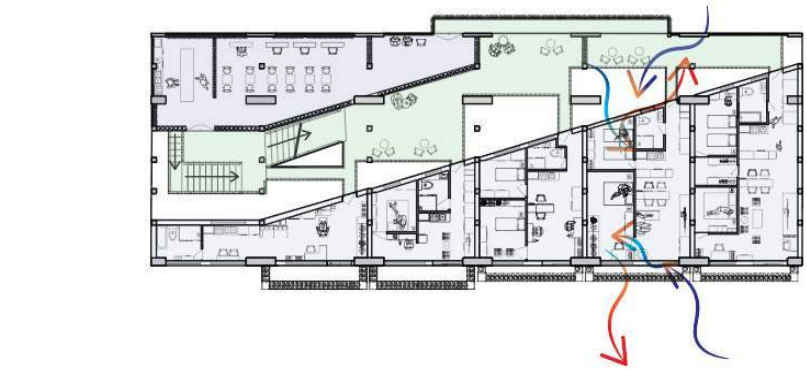


Figure 67

# Design relation




Figure 68



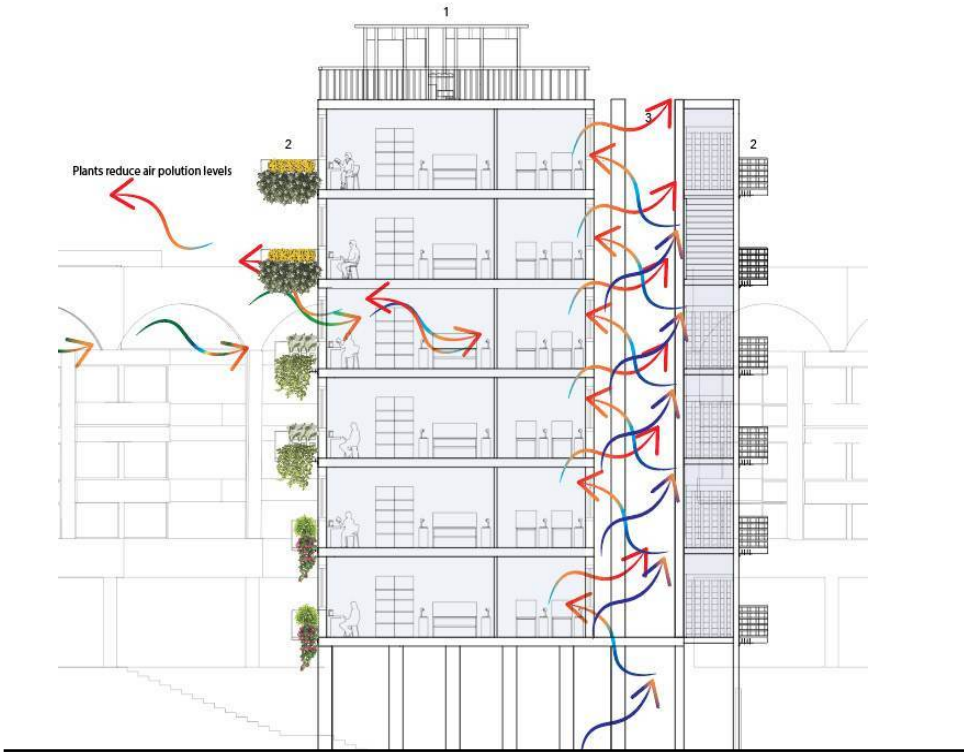



Figure 69

References

Figure 1: personal drawing
Figure 2: personal drawing
Figure 3: https://blog.frontiersin.org/2016/01/12/frontiers-in-built-environment-publishes-its-first-ebook/
Figure 4: http://expeditieaarde.blogspot.com/2011/11/koppen-climate-classification-system.html
Figure 5: Personal drawing
Figure 6: Personal drawing
Figure 7 https://blog.frontiersin.org/2016/01/12/frontiers-in-built-environment-publishes-its-first-ebook/
Figure 8: Personal drawing
Figure 9: Personal drawing
Figure 10: Personal drawing
Figure 11: http://annacfarrell.com/delta-air-lines
Figure 12: http://sitraka.co.uk/the-photon-shower/
Figure 13: http://sitraka.co.uk/the-photon-shower/
Figure 14: personal work
Figure 15: personal work
Figure 16:https://vimeo.com/80947343
Figure 17: personal drawing
Figure 18: personal drawing
Figure 19 https://clipartion.com/free-clipart-5398/
Figure 20: personal work
Figure 21: personal work
Figure 22: http://passivesolar.sustainablesources.com/
Figure 23: https://www.tealproducts.com/natural-ventilation-control
Figure 24: personal work
Figure 25: personal work
Figure 26: personal work
Figure 27: lecture by Rosa Schiano-Phan
Figure 28: personal drawing
Figure 29: personal drawing
Figure 30: https://www.echotape.com/blog/contractors-field-guide-building-envelope/
Figure 31: personal photo
Figure 32: Tugsce Yigit's drawing
Figure 33: personal photo
Figure 34: https://itunes.apple.com/gb/app/aftab-luminance/id1054568119?mt=8
Figure 35: https://itunes.apple.com/gb/app/aftab-luminance/id1054568119?mt=8
Figure 36: https://www.semanticscholar.org/paper/ADAPTIVE-THERMAL-COMFORT-AND-SUSTAINABLE-THERMAL-Nicol-Humphreys/346a9f17076b728357e679e6c7e2216cd9681e16
Figure 37: https://www.semanticscholar.org/paper/ADAPTIVE-THERMAL-COMFORT-AND-SUSTAINABLE-THERMAL-Nicol-Humphreys/346a9f17076b728357e679e6c7e2216cd9681e16
Figure 38: http://buildingcodenyc.com/sketches/exit-door-width-capacity/
Figure 39:personal work

Figure 40: http://www.estechja.com/building-management-solutions/
Figure 41: http://www.futurebyenergy.co.uk/epc/
Figure 42: https://www.researchgate.net/publication/258140202_A_Field_Study_on_Thermal_Comfort_of_Occupants_and_Acceptable_Neutral_Temperature_at_the_National_Museum_in_Malaysia/figures?lo=1

Figure 43: https://www.edie.net/news/10/In-charts--The-UK-s-record-breaking-year-for-low-carbon-generation/
Figure 44: personal photo
Figure 45: personal photo
Figure 46: personal photo
Figure 47: personal photo
Figure 48: personal photo
Figure 50: personal drawing
Figure 51: personal drawing
Figure 52: personal photo
Figure 53: personal photo
Figure54: https://www.landlord-certificates.co.uk/epc/
Figure 55: https://en.wikipedia.org/wiki/1998_Greater_London_Authority_referendum
Figure 56: https://www.researchgate.net/publication/305904072_Identification_of_Research_Areas_to_Advance_the_Adoption_of_Cross-_Laminated_Timber_in_North_America/figures?lo=1&utm_source=google&utm_medium=organic

Figure 57: https://theconstructor.org/question/volume-of-concrete-sand-cement-aggregate/

Figure 58: ttps://www.thetimes.co.uk/article/pollution-destroying-life-of-englands-chalk-streams-knvm-rx87m
Figure 59: ttps://www.thetimes.co.uk/article/pollution-destroying-life-of-englands-chalk-streams-knvm-rx87m
Figure 60: personal drawing
Figure 61: personal drawing
Figure 62: personal photo
Figure 63: personal photo
Figure 64: personal photo
Figure 65: personal photo
Figure 66: personal photo
Figure 67: personal photo
Figure 68: personal drawing
Figure 69: personal drawing

All work produced by Sebastian Victor Dawber

---

https://www.sebastianvictordawber.co/

---

This is work produced by the parties mentioned anyone using the work without agreement signiture and physical represented has done so in their own doing without the parties mentioned

---

Https://www.instagram.com/sarc_studio/

---

Documents by Sebastian Dawber

Dissertation
Research into ..



https://www.sebastianvictordawber.co

Created for ...
BSc Architecture and environmental design
Undergraduate  RIBA ARB part I

All work produced by Sebastian Victor Dawber

2018 – 2019

This diary summarises a year of technical environmental study, covering climate analysis, passive design, ventilation, daylighting, thermal comfort and energy systems. It documents how lectures, fieldwork and industry visits were applied to live design work, and can serve as a reference for environmental design strategies in future projects.



Documents by Sebastian Dawber
https://www.sebastianvictordawber.co

Created for ...
BSc Architecture and environmental design
Undergraduate RIBA ARB part I

All work produced by Sebastian Victor Dawber

https://www.sebastianvictordawber.co/

This is work produced by the parties mentioned anyone using the work without agreement signiture and physical represented has done so in their own doing without the parties mentioned

Https://www.instagram.com/sarc_studio/